

Graph Neural Networks for Decentralized Multi-Agent Reinforcement Learning in Power-Grid Topology Control

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Environment Grid2Op / L2RPN · built on MARL2Grid

Can a decentralized graph policy trained on a 14-bus grid still control a 36-substation grid?

IEEE 14-bus → L2RPN WCCI 2020

The argument in four steps

Topology control is nearly free flexibility — but every published policy is welded to the grid it was trained on.

1

The problem

Almost-free flexibility.
Combinatorial search.
Regional operation.

2

The design

State and action interface
both independent
of grid size.

3

The evidence

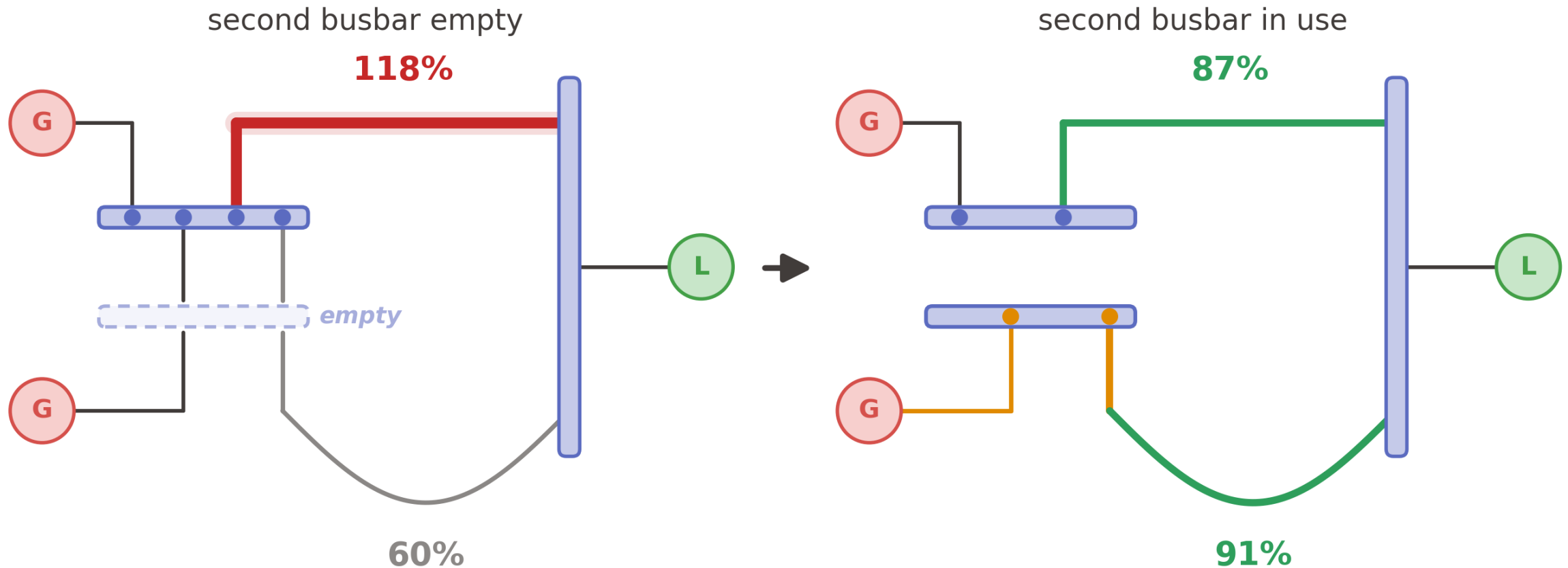
bus14 → WCCI36.
Zero-shot, from scratch,
fine-tuned.

4

The answer

Incomplete — but 3× do-
nothing,
and better than retraining.

How a topology action relieves an overload

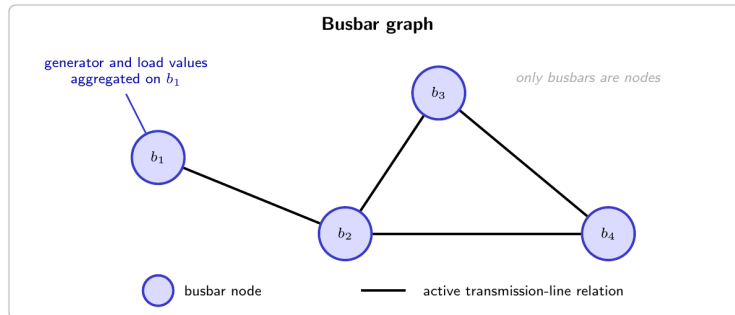


Same lines, same generation, same demand. **Only the switch positions changed.**

The problem, in three steps

Almost-free flexibility · an action space too large to search · a grid already operated region by region.

1 · TOPOLOGY CONTROL



Reconfigure busbars, not generation.

Marginal cost ≈ 0 .

Today: a short offline list.

2 · TOO BIG TO SEARCH

178 → 66,965

bus14 → WCCI36

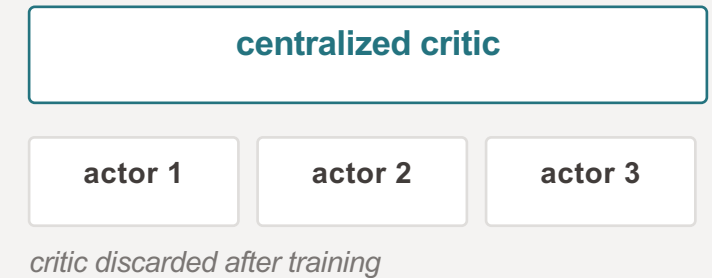
unitary topology actions

2^d assignments per substation.

Cooldowns couple decisions in time.

→ Reinforcement learning.

3 · ALREADY REGIONAL



One control centre per region.

A central actor grows with the grid.

CTDE: global critic, training only.

State of the art: four families, one shared limitation

Existing methods shrink the search or coordinate centrally. None produces a policy you can move to another grid.

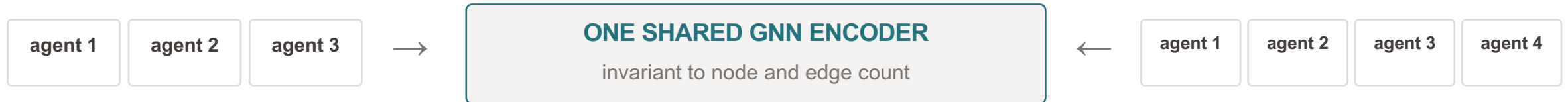
Family	Representative work	What it does	Why it stays put
Single-agent	SMAAC · Subramanian 2021	Afterstate actor–critic, hierarchical where / how	Action space reduced or factorized per grid
Policy + heuristics	PowRL	Act only when loading is high; overload manager	The rule sits outside the policy
Hierarchical / centrally coordinated	Manczak · van der Sar · de Mol	Split when / where / how; central arbitration	A component still sees the whole grid
Fully decentralized	MARL2Grid (starting point)	Local observations, no communication, no selector	Fixed-width actors welded to one grid

The gap. A graph encoder frees the **state** from grid size. The **action interface** — one output row per action index — does not.

One encoder, two levels of sharing

A GNN is size-invariant — and regions of comparable size make the same encoder work across grids too.

bus14 · 3 regions



1 same weights for every agent

2 same weights on the other grid

Regions are the same order of size — 4–6 substations on bus14, 5–17 on WCCI36 — so the encoder always reads a local graph, never a whole grid of 14 against 36.

THE STATE TRANSFERS

One weight set reads any local graph, on either grid.

→ **the bus14 encoder loads on WCCI36 unchanged**

THE ACTION HEAD DOES NOT

One output row per action index, and $|A|$ differs by agent and by grid.

→ **score actions in a way independent of $|A|$**

Third condition, separately: the same channel must carry a comparable number on both grids.

PART 2

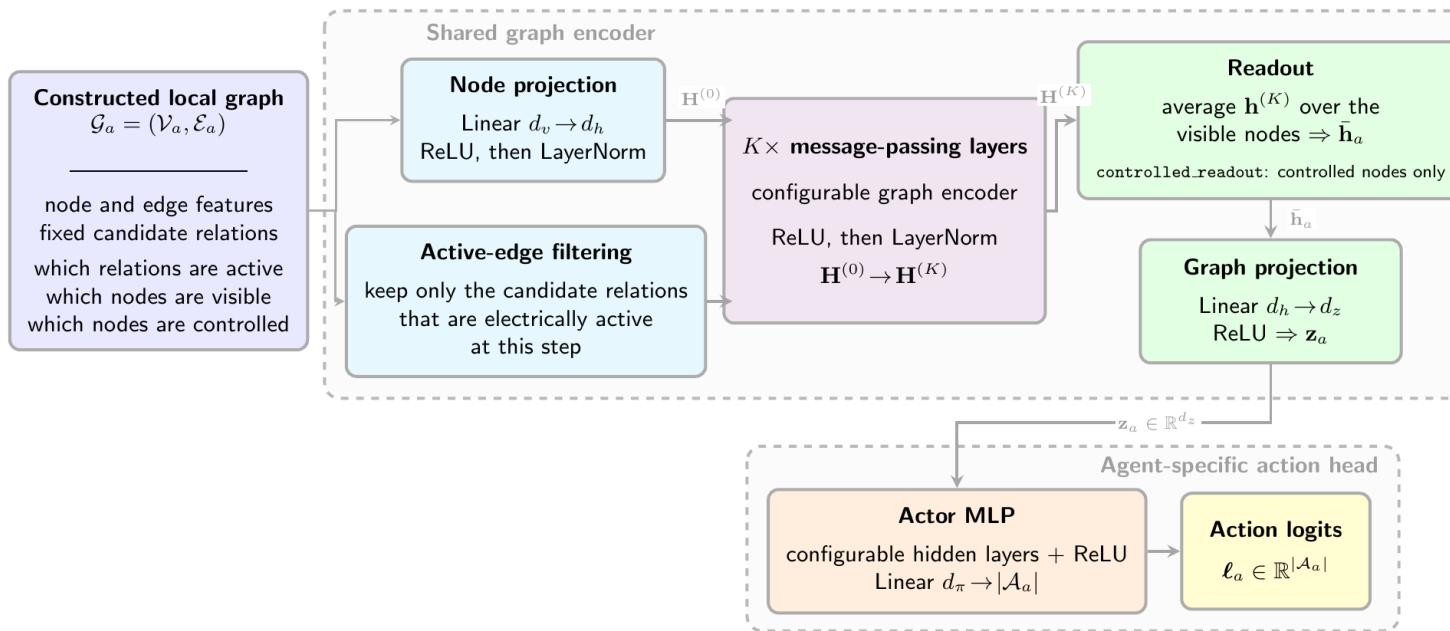
Designing a policy that can move

The two mechanisms that let one set of weights read a different grid — and the input transformation that makes the numbers comparable.

1. The problem and the gap
- 2. Designing a policy that can move**
3. Does it cross grids?
4. What it means

Method I — one shared encoder, one local graph per agent

The state stops depending on grid size: the same weights read any local neighbourhood.



Local graph $\rightarrow$ active-edge filtering $\rightarrow$ shared GINE encoder $\rightarrow$ readout $\rightarrow$ agent-specific head.

What the graph and the encoder can be: next two slides.

Local graph

Own substations and every line touching them.
Across a boundary: the shared line only.

Dynamic masking

Fixed tensor shape; inactive edges dropped before message passing.

Shared weights

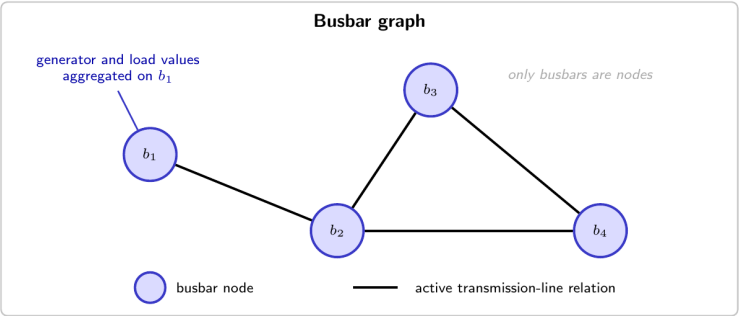
One encoder for every agent; only the action head is agent-specific.

Centralized critic

A separate MLP over the global state, discarded after training.

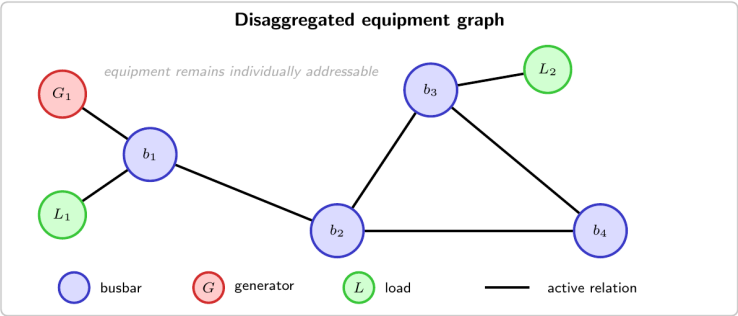
Three graph representations

How much of the substation the encoder actually sees — and what that leaves visible across a regional boundary.



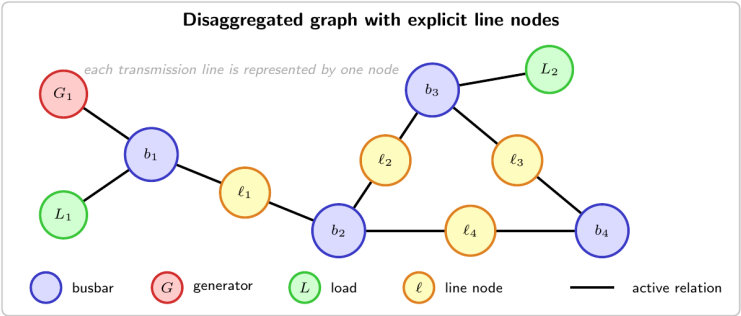
Busbar

one node per busbar; power summed onto it



Disaggregated

generators and loads become their own nodes



Line-node

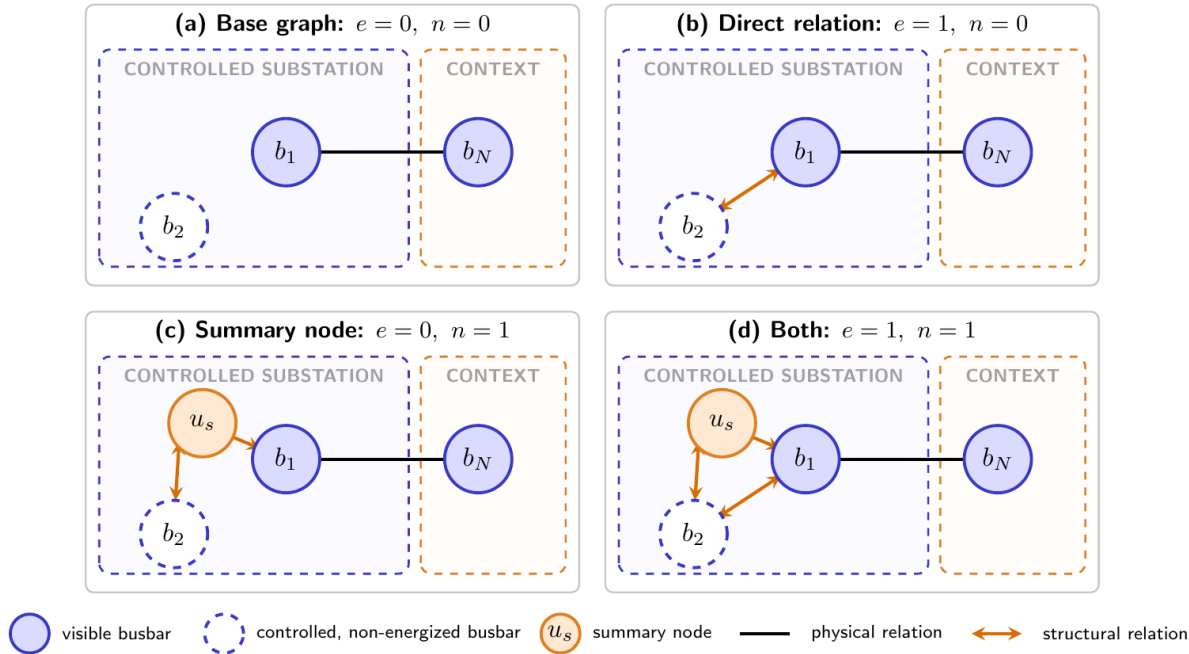
every transmission line becomes a node

What a local actor sees of its neighbour	
Busbar	the shared line and the far-end busbar, with its aggregated power and angle
Disaggregated	the shared line and the far-end busbar — but no neighbouring generator or load
Line-node	the shared line node only — no neighbouring busbar, equipment or attachment

The line-node graph is the one carried into transfer: a boundary line stays addressable without constructing the neighbour's busbar.

Three optional augmentations

Extra structure, zero extra measurement — and never extended into a neighbouring region.



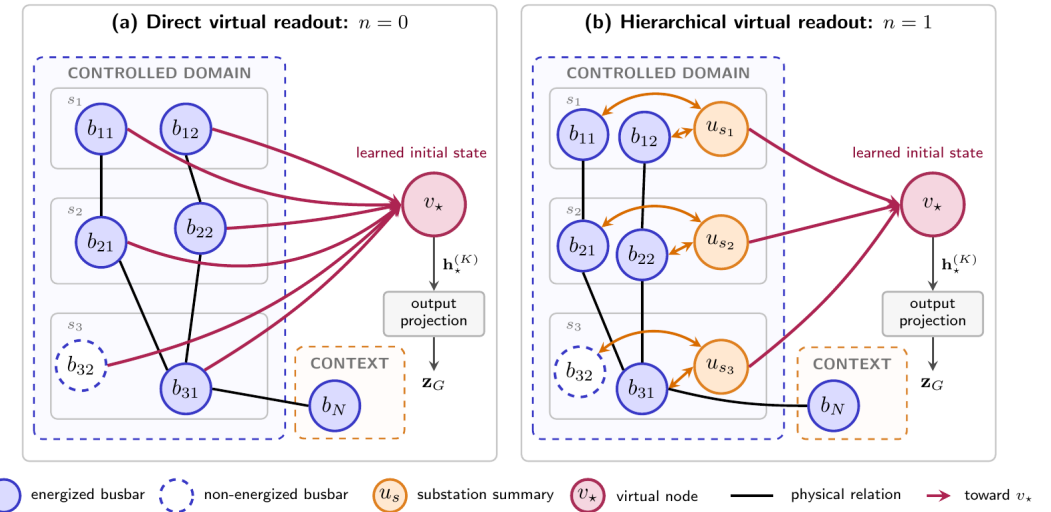
e direct same-substation edge

n summary node

The two busbars of a substation exchange messages in one hop, or in two through a learned hub.

The hub is initialised from structure only, not identity.

Three binary factors — the 2^3 cells of Screen D.



v virtual-node readout

One learned node collects from the controlled busbars and replaces the readout entirely.

Edges point one way: it cannot send back into the graph.

What the encoder can be

Message-passing operator, capacity, and readout — the three knobs settled by Screens A, B and C.

OPERATOR screen A

GINE vs **GAT**

$$m_{u \rightarrow v} = \text{ReLU}(h_u + W_e e_{uv})$$

Edge features enter every message.

CAPACITY screen B

	16	32	64	128
K=1				
K=2				
K=3				

layers × width **selected K = 2, width 128**

READOUT screen C

which nodes	mean	max
all visible	.	.
controlled	.	.
energized	•	•
controlled ∩ energized	.	.

8 readouts or one virtual node instead

CARRIED FORWARD INTO EVERY LATER EXPERIMENT

GINE · K = 2 layers · width 128 · readout over energized nodes · no augmentation

Screen results on slide 14. Depth, width, readout and augmentation heatmaps in the backup slides.

Method II — score each action from the nodes it touches

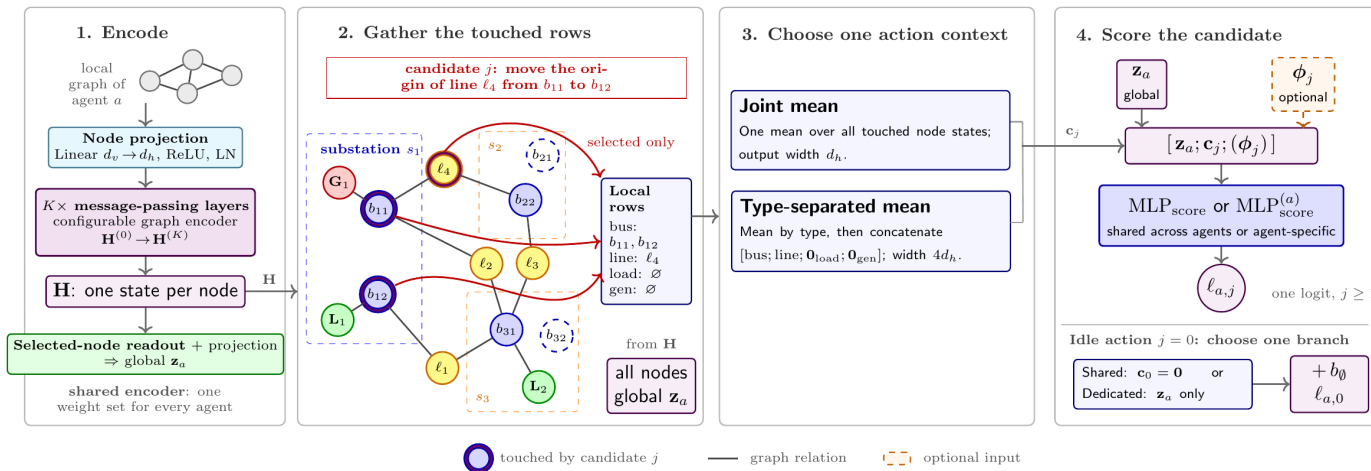
The piece missing from the literature: the head stops depending on the number of actions.

DEFAULT HEAD

$$\ell_a = \text{MLP}_a(z_a) \quad \text{one row per action index}$$

CANDIDATE SCORING

$$\ell_{a,j} = \text{score}(z_a, c_j, \phi_j) \quad \text{one shared scorer}$$



Why it transfers

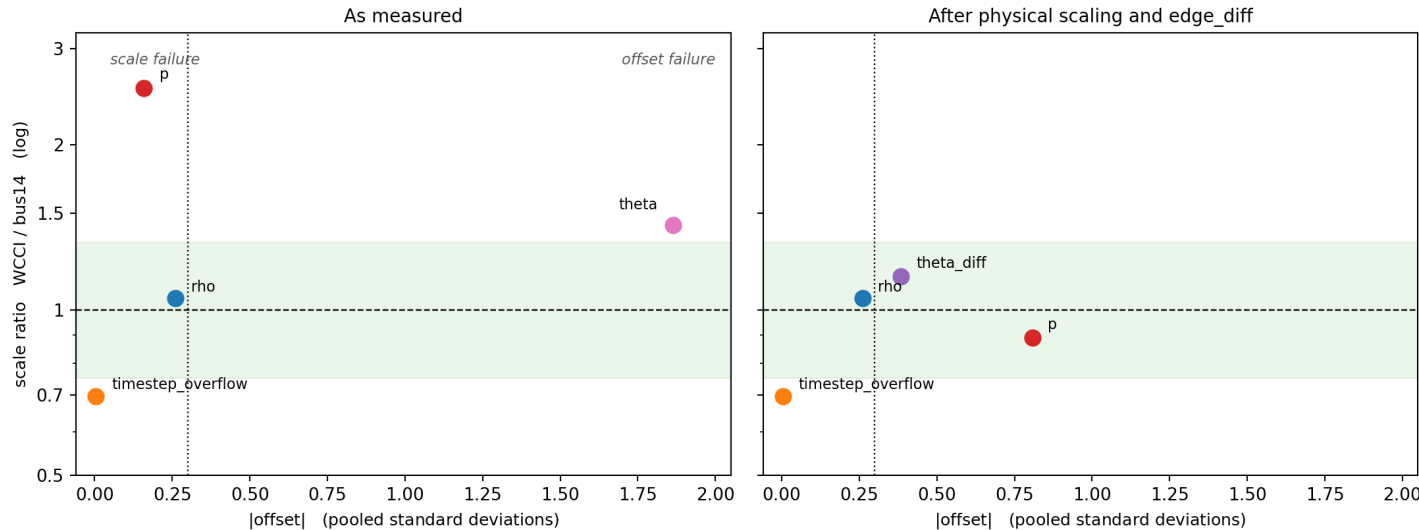
Parameters do not grow with $|A|$. Actions are compared through the graph nodes they affect.

Three choices $\rightarrow$ 8 variants

Pooling: joint or typed mean · descriptor on / off · idle action: shared or dedicated head.

Making a feature mean the same number on both grids

A shared encoder is worthless if a channel learned on bus14 arrives on a different scale or reference.



Offset against scale ratio, per node feature. Bottom-left is compatible. 426k bus14 and 1.24M WCCI node observations.

NL original inputs

NLS power ÷ largest generator, absolute angles $\rightarrow \Delta\theta$

p already transfers

Ratio 1.05 — divided by its own thermal limit.

Angles were the worst

Offset +1.86 SD, KS 0.678 — different slack buses.

Fix: line-angle difference $\Delta\theta$

No reference bus. KS 0.678 $\rightarrow$ 0.230.

Power is a real trade-off

Spread 2.54 $\rightarrow$ 0.89, but KS 0.201 $\rightarrow$ 0.511.

Standardization cannot align

Frozen source statistics leave KS unchanged.

PART 3

Does it cross grids?

One source grid, one target grid, one fixed evaluation protocol — and four ways of getting a policy onto the target.

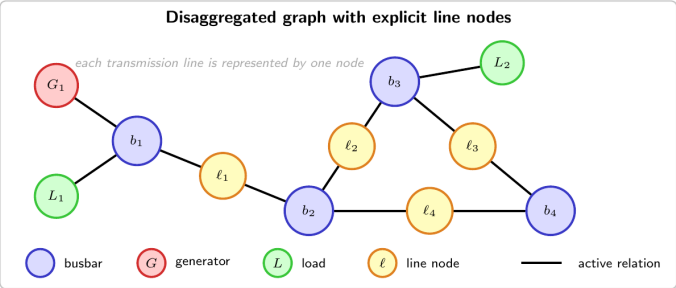
1. The problem and the gap
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The experimental frame

Overall survival hides the cases that matter. The evidence is the 24 chronics where doing nothing already fails.

	bus14 source	WCCI36 target
Substations	14	36
Transmission lines	20	59
Decentralized agents	3	4 (5, 5, 9, 17 substations)
Chronics	1,004	2,880
Unitary actions	178	66,965 → capped at k
Maintenance	none	disabled

Maintenance is off so a transferred policy is not tested on a hazard absent from its source grid: do-nothing survival moves 27.3% → 57.3%.



The line-node graph used for transfer.

Mean survival

Survived steps ÷ maximum episode length.

50 fixed WCCI chronics

Do nothing completes 26 — easy. The other 24 are the difficult cohort.

Greedy is a reference, not a ceiling

It queries the simulator at every step.

56.0%

do nothing, all 50

8.34%

do nothing, 24 difficult

The number every controller here is trying to beat.

On the source grid, the design questions are settled

Six controlled screens on bus14. The best actors — including the transferable one — exceed 99% mean survival.

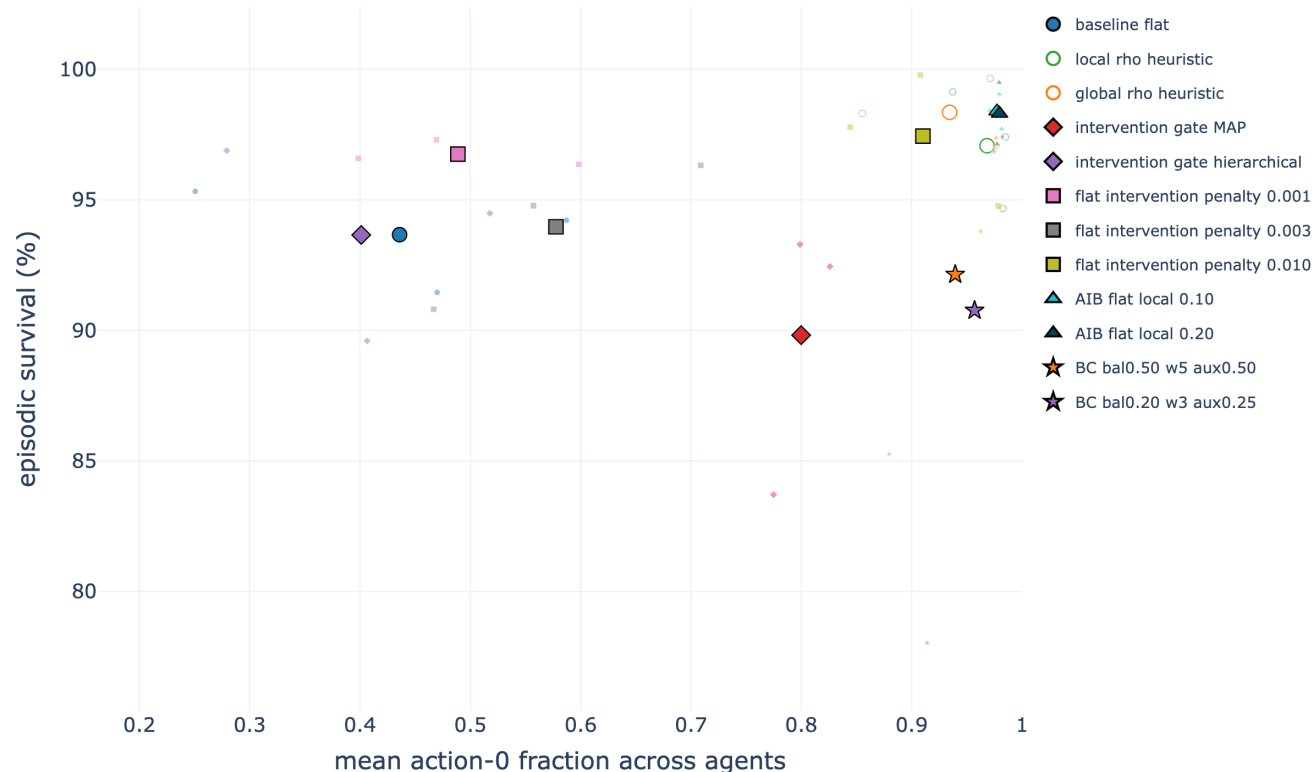
Screened factor	Result	Carried forward
A Message-passing operator	GINE 94.72 vs GAT 91.06	GINE
B Depth × width (3 × 4)	No monotone rule; mp2_h128 converges first and holds	K = 2, width 128
C Readout scope × reducer (4 × 2)	Energized nodes lead: 99.39 max, 98.93 mean	energized / max
D Structural augmentations (2 ³)	None earns its cost; virtual readout collapses to 18–77	unaugmented e0n0v0
E Asset message direction (3 × 3)	gen a2b + load b2a 99.12; bidirectional 99.02	both kept
F Candidate head (2 ³)	6 of 8 cells ≥ 98.2, best 99.52; no factor dominates	all 8 kept

Two branches leave this chapter. A–E keep a fixed-index head → only the **encoder** transfers. F keeps the whole interface → the **complete policy** crosses grids, at no source-grid cost.

One seed per cell; best checkpoint scored on all 201 held-out bus14 chronics.

...and the controller can learn to act rarely

A policy that reconfigures at every opportunity is unusable in a control room. Sparsity can be learned, not imposed.



Mean survival against the fraction of do-nothing decisions.

FIVE MECHANISMS TESTED

- Evaluation-time p overrides
- Learned intervention gate
- Fixed non-idle penalty
- Adaptive intervention budget (AIB)
- Behavioural cloning

Local-safe AIB wins on both axes

Each agent adapts its own penalty from the loading it can see.

93.7 → 98.4%

mean survival

56.6 → 2.1%

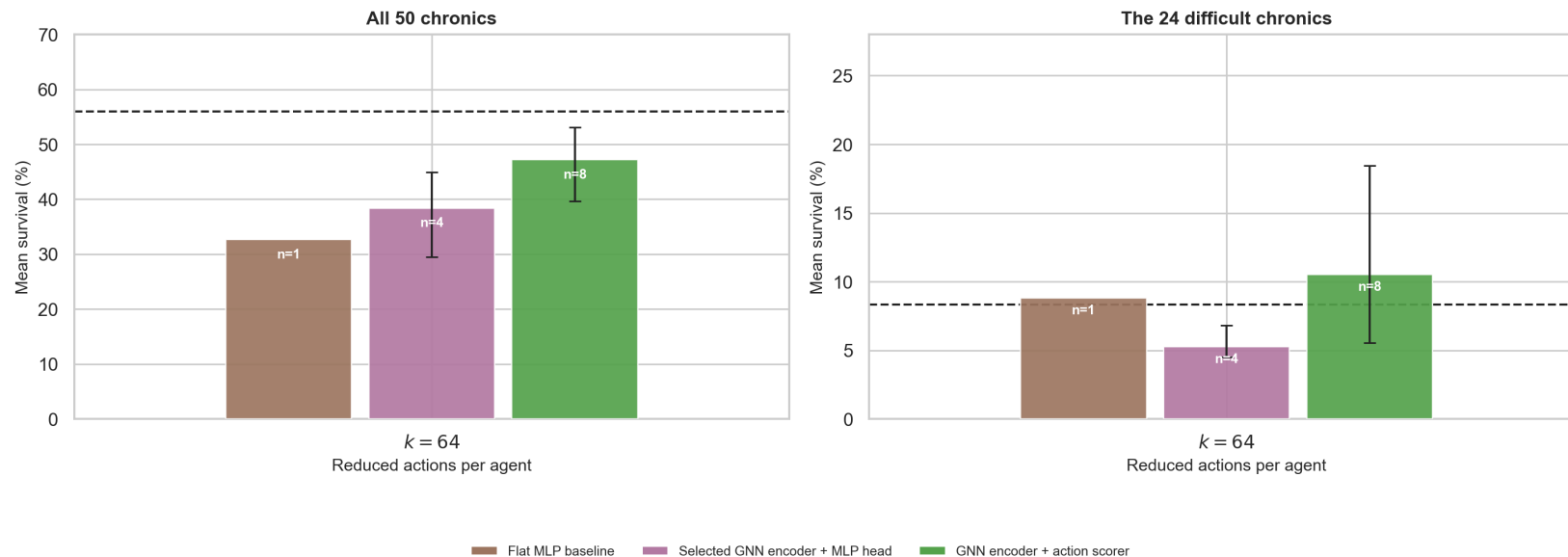
non-idle actions

The same local- p idea returns at transfer time as a safety gate.

First surprise: retraining on the target does not solve it

Every model family retrained directly on WCCI stays below do-nothing overall — including the graph actors.

Matched WCCI controls trained from scratch at $k = 64$



Matched controls trained from scratch at $k = 64$. Bars are family means, whiskers min-max.

Overall survival is the wrong headline. It is dominated by the 26 easy chronics do-nothing already completes.

ALL 50 CHRONICS

Flat MLP 32.6%

GNN + fixed head 38.4%

GNN + action scorer 47.2%

Do nothing 56.0%

THE 24 DIFFICULT CHRONICS

GNN + fixed head 5.3%

Do nothing 8.34%

Flat MLP 8.8%

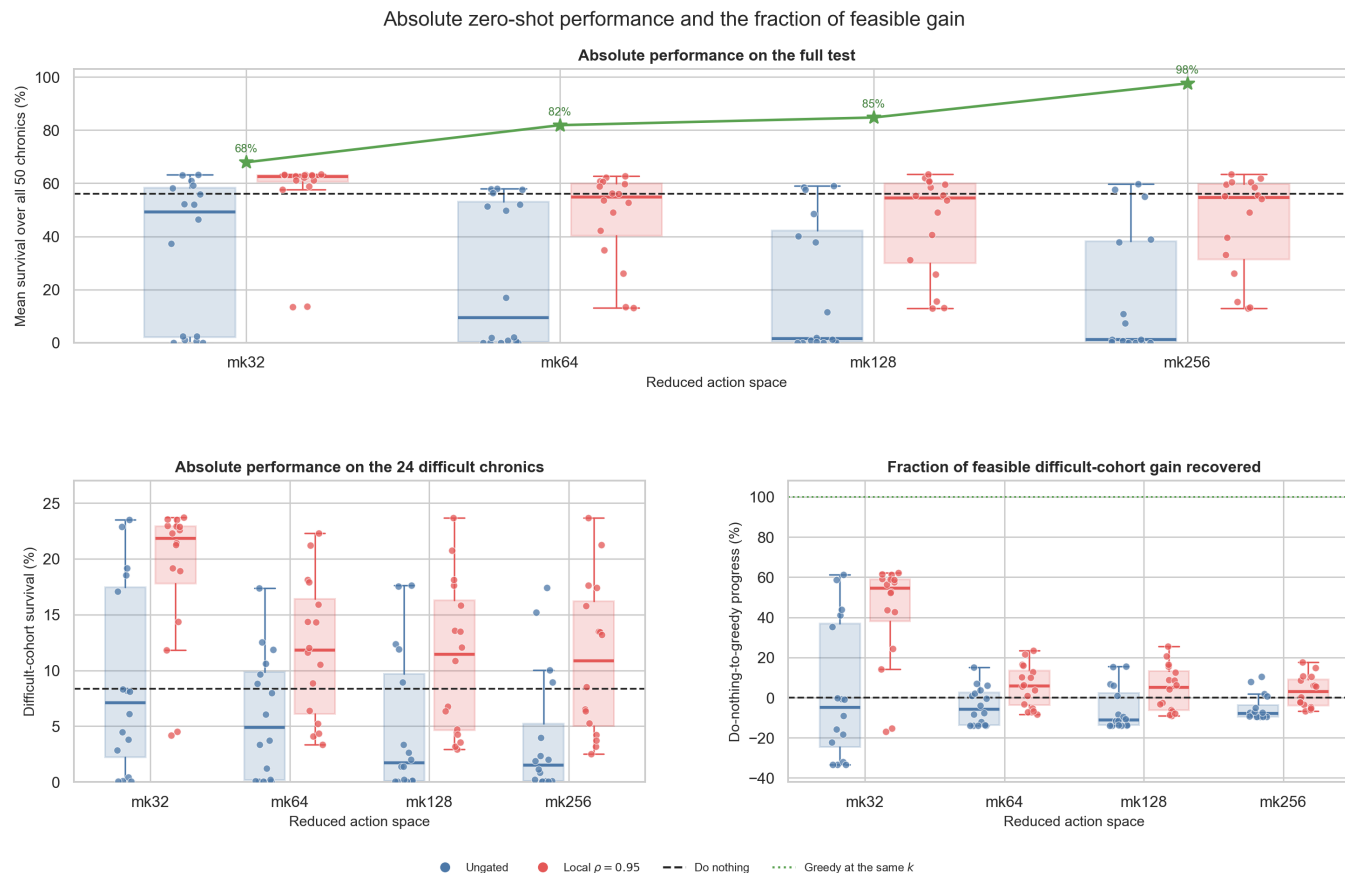
GNN + action scorer 10.5%

its best cell 18.5%

The ordering flips. Only the candidate scorer is clearly above do nothing here.

Zero-shot: the bus14 actor does control WCCI — conditionally

It works with a small action set and a local safety gate. More actions make the policy worse, not better.



Gated, $k = 32$

62.5% overall, 21.9% difficult — 54.6% of the gain greedy makes available.

Then it collapses

5.8 · 5.2 · 2.9% recovered at $k = 64, 128, 256$ — while greedy climbs 68 → 98%.

Ungated stays below do nothing

At every action-space size.

Selection is the constraint

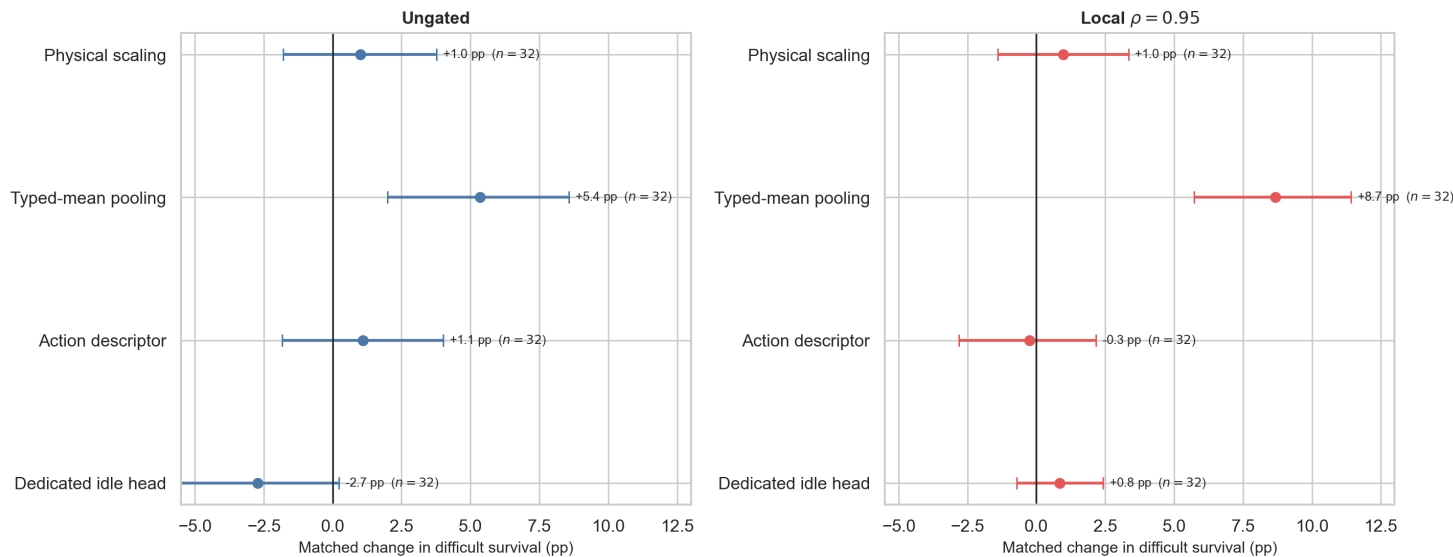
Not availability. A longer list is harder to rank.

16 architecture cells per box. Green stars: greedy at the same k . Dashed: do nothing. Gate blocks an agent while its own maximum loading is below 0.95.

What actually survives the change of grid

One architectural choice transfers clearly: keeping the physical roles of the touched nodes separate.

Architecture effects depend on the gate condition



Matched change in difficult-cohort survival, one factor at a time. Bars are bootstrap intervals over 32 matched pairs.

Typed-mean pooling

+5.4 pts ungated, +8.7 gated. Bus, line, load and generator averaged separately.

Physical scaling

≈ +1 pt — less than the input analysis promised.

Descriptor ≈ 0, idle head negative

-2.7 pts ungated, neutral gated.

DECISION-TIME CHANGES

Gmax-delta +4.6 pts · adaptive budget +6.0 pts.

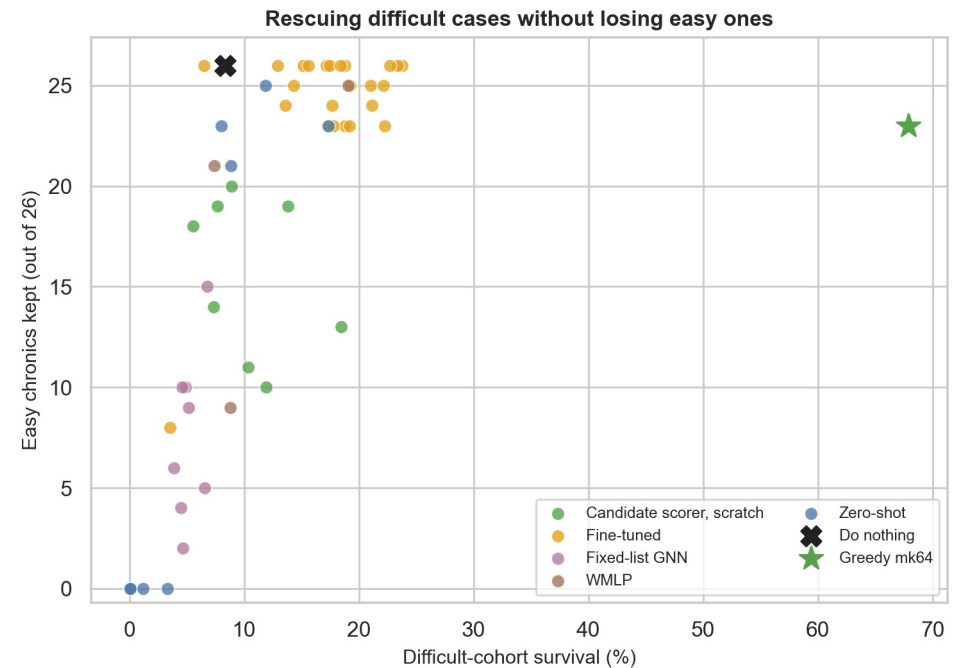
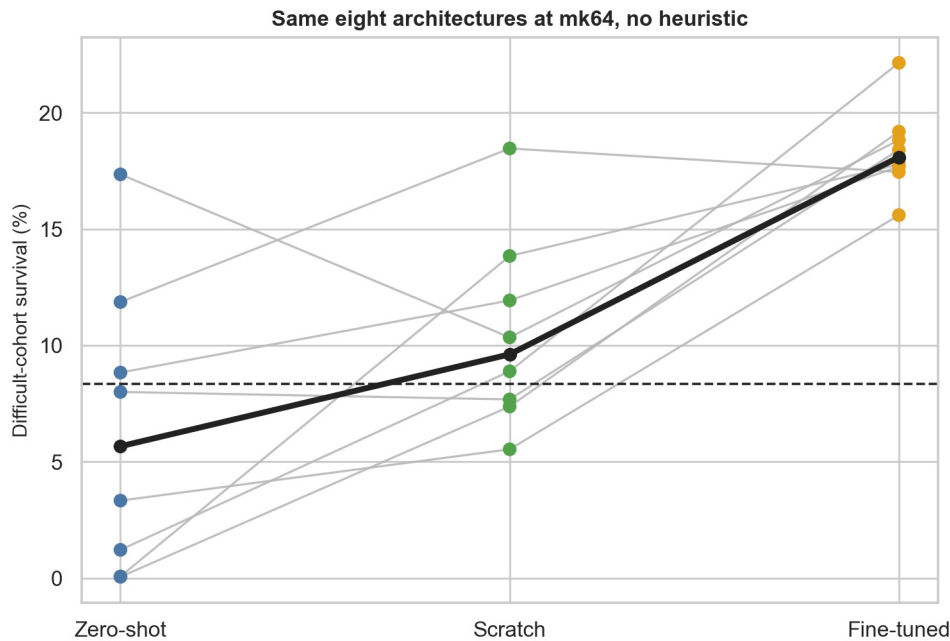
THE GATE IS NOT A UNIVERSAL SAFETY LAYER

+7.2 pts zero-shot, but -2.2 on the scratch scorer and -2.8 on the flat MLP. Validate per policy.

Source initialization is worth more than target training

Fine-tuning beats zero-shot in 8 of 8 cells and scratch in 7 of 8 — on a third of the budget.

Transfer becomes reliable after target-grid fine-tuning



Median difficult-cohort survival

5.7% zero-shot → 9.6% scratch → 18.1% fine-tuned.

Matched gains

+12.0 pts over zero-shot, +7.9 over scratch.
≈5M target steps against 15M.

The trade-off that matters

Upper-right corner: rescue difficult chronics without losing easy ones.

Still open: does the encoder alone carry the transfer?

The complete-policy branch is done; the encoder-only branch is still running.

THE DESIGN IS FIXED

Freeze the bus14 encoder, discard its action heads, learn new WCCI heads. Four representations $\times$ four routes:

- **Target scratch** — random encoder, all trainable
- **Random frozen** — random encoder, heads only
- **NL frozen** — bus14 checkpoint, original inputs
- **NLS frozen** — bus14 checkpoint, scaled inputs + $\Delta\theta$

Random-frozen is the control. New heads can learn even from arbitrary features.

REQUIRED BEFORE THE DEFENCE

Outputs

- Matched learning curves, four routes
- 4×4 difficult-cohort endpoint table
- Overall survival, rescues, easy chronics kept

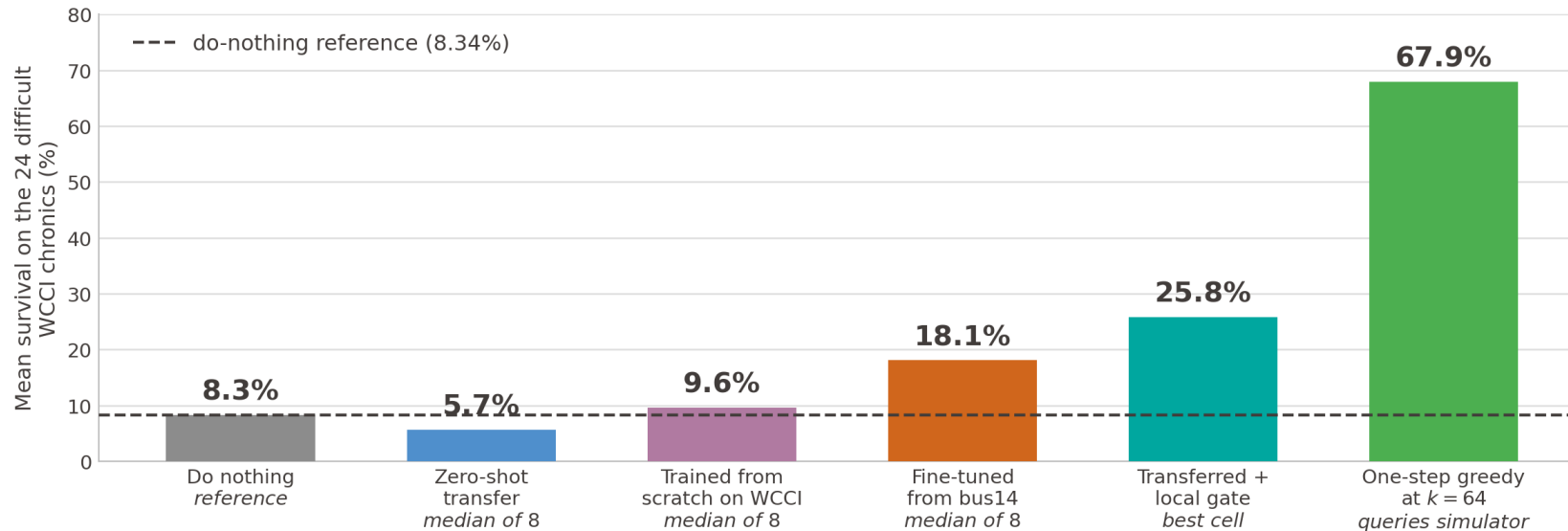
Protocol fields still pending

- Action cap and dataset path
- Target training budget
- Seed set
- Head optimizer and learning rate
- Checkpoint-selection score
- Gate condition at evaluation

The interpretation rules are already fixed in the chapter, so the conclusion cannot be chosen after seeing the data.

What crosses, and what does not

Transfer is incomplete — but about three times do-nothing, and better than retraining the same actor from scratch.



What transferred

The candidate-action interface. 21.9% on the difficult cohort at $k = 32$ with the gate.

What did not

Ranking large action sets: 55% of feasible gain at $k = 32$, 3% at $k = 256$.

What is still uncertain

One seed, 50 chronics. Matched differences hold; small cross-block gaps do not.

Limitations and where this goes next

Limitation	What it means for the claims	Next step
One seed per cell	Intervals describe architecture spread, not training runs.	Repeat the transfer block over seeds.
Two grids, one direction	Nothing tests the reverse direction or a third topology.	Add a third grid; is the k = 32 advantage general?
Unequal training budgets	Comparisons hold inside a block, not across blocks.	Re-run cross-block comparisons at one budget.
Static offline action reduction	Candidates ranked once; the policy never revisits the list.	Curriculum or state-dependent candidate sets.

The design conclusion is more robust than any single number. Portable interface, separate node roles, per-policy gate, conservative fine-tuning.

Thank you

Questions welcome — supplementary slides follow.

Supervisors: Prof. Fink · Prof. Peng · Dr. Nejjar

NUMBERS TO HAVE READY

Do nothing, 24 difficult **8.34%**

Best bus14 actor **99.5%**

Scratch on WCCI, difficult **9.6%**

Zero-shot median, difficult **5.7%**

Fine-tuned median, difficult **18.1%**

Best transferred + gate **25.8%**

Greedy at $k = 64$, difficult **67.9%**

Typed-mean pooling effect **+8.7 pts**

Local gate on zero-shot **+7.2 pts**

Angle KS, raw $\rightarrow \Delta\theta$ **0.678 $\rightarrow$ 0.230**

BACKUP

Supplementary material

Protocols · complete screening results · feature-shift analysis · full transfer matrices · reproducibility.

Evaluation protocol and checkpoint selection

Every reported endpoint is a checkpoint selected on training-split rollouts, then scored once on a held-out set.

Campaign	Environment and budget	Checkpoint selection	Statistical scope
Stable MLP baseline	bus14; 15M steps; 20 parallel envs	Periodic deterministic 20-chronic training-split rollouts; keep the highest rollout mean	80/20 chronic split; multi-seed core comparisons
Screens B–F	bus14; 15M steps; 72 envs, 576 rollout steps	Evaluation every 82,944 steps on a changing 20-chronic training rollout; the selected checkpoint is then scored on all 201 held-out chronics	One seed per factorial cell (seed 0)
Input-distribution study	bus14 and WCCI; 6,000 do-nothing steps; 300-step cap per chronic	No learned checkpoint; every visible local-graph node is recorded	426,000 and 1,236,000 node observations
WCCI transfer and adaptation	bus36_wcci_nomaint; zero-shot, 15M-step scratch, or ≈5M-step fine-tuning	Zero-shot uses the selected source checkpoint; WCCI-trained actors use their final checkpoint	Seed 0; the same 50 chronics — 26 easy, 24 difficult

Mean survival for one chronic is the number of survived time steps divided by that chronic's maximum episode length; the reported value is the unweighted mean over the evaluated chronics.

Shared MAPPO settings and the tuned MLP baseline

Screens B–F share one optimization recipe so each factorial changes only the named architectural choice.

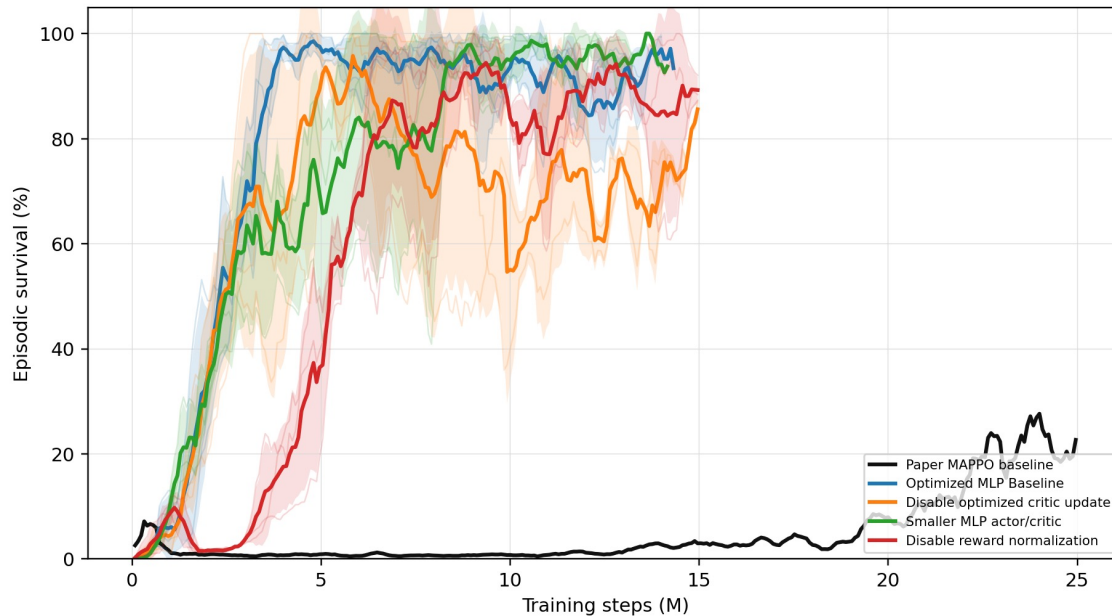
Component	Fixed setting (Screens B–F)
Optimization	10 update epochs, 16 minibatches, LR 1e-4 with linear decay, max gradient norm 1.0
Returns and PPO	$\gamma = 0.9$, GAE $\lambda = 0.95$, clip ratio 0.2, target KL 0.02, entropy coefficient 0.01
Actor and critic	Shared graph encoder with two 128-unit action-head layers; centralized 3×256 MLP critic
Preprocessing	Normalized flat observations and rewards; graph physical scaling and sparse control disabled during the screens
Exploration	No initial do-nothing prior, no action-0 logit bonus

Hyperparameter	Tuned	MARL2Grid paper
Parallel environments	20	10
Total environment steps	15,000,000	25,000,000
Actor learning rate	1e-4	3e-5
Critic learning rate	1e-4	3e-5
Discount γ	0.9	0.99
Update epochs	10	80
Minibatches	16	4
Max gradient norm	1.0	10.0

Reward normalization and one critic update per batch (instead of one per agent) are the other two changes; together they take the flat MLP baseline to 93.4% final / 98.6% peak mean survival on bus14.

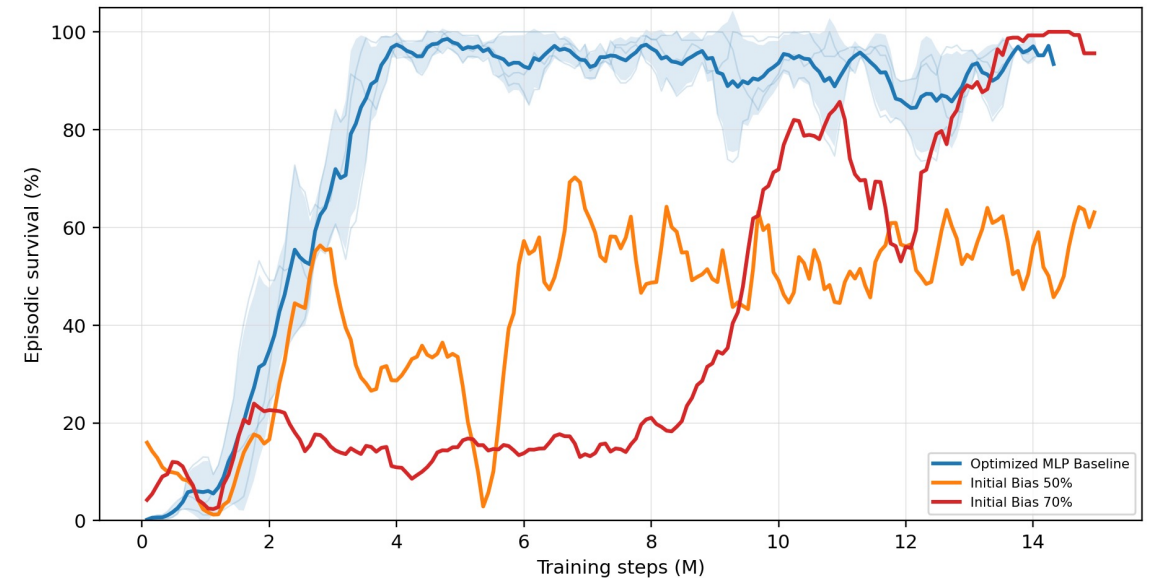
Flat MLP baseline: core ablations and do-nothing bias

The reference is chosen for its convergence trajectory, not for the largest isolated endpoint.



All core MLP baseline rerun survival curves on one axis.

Run	Final (%)	Peak (%)
Optimized MLP baseline	93.4	98.6
Disable optimized critic update	93.1	95.8
Smaller MLP actor/critic	93.7	100.0
Disable reward normalization	81.8	97.1
Initial bias 50%	50.8	70.2
Initial bias 70%	99.3	100.0



Optimized baseline against initial do-nothing bias 50% and 70%.

A 70% do-nothing bias finishes highest but learns slowly; a 50% bias is unstable. The zero-bias actor reaches the high-survival regime earliest and holds it, so it is the reference used downstream.

Sparse intervention: the five mechanisms

Each mechanism acts at a different point — training objective, action factorization, or evaluation-time override.

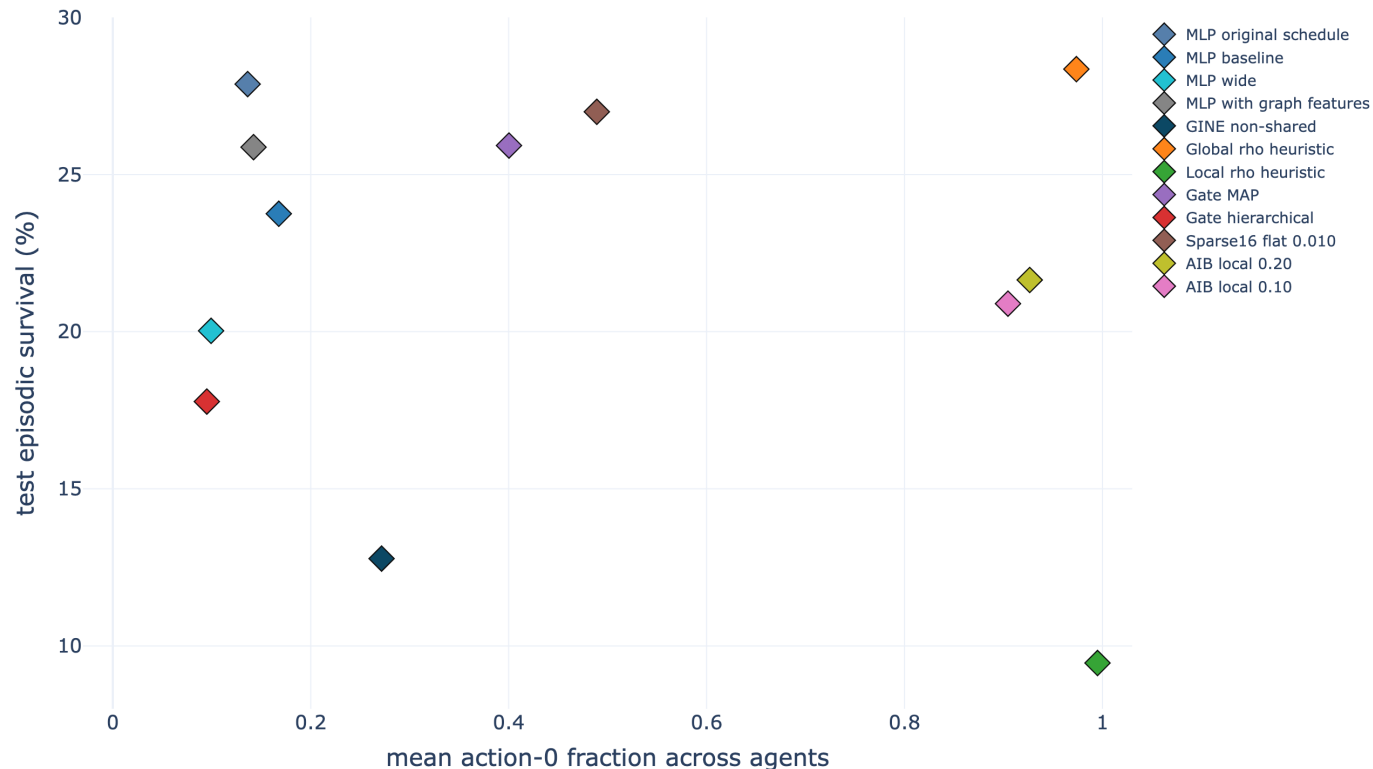
Method	Training effect	Evaluation effect	Main risk
Baseline	Samples from the flat policy	Greedy argmax by default	No explicit mechanism favouring sparse control
p heuristic	No training change	Hard no-op override in safe states	Evaluation is manually changed
Intervention gate	Changes the action factorization and sampling	Final-action MAP or hierarchical greedy	Gate collapse can starve non-idle exploration
Fixed penalty	Fixed reward penalty on non-idle actions	No action override	The penalty coefficient must be tuned
AIB local-safe	Adaptive state-dependent reward penalty	No action override	Depends on the local-p safety-weight design
Behavioural cloning	Distils the heuristic into actor weights	No action override	Struggles with extreme action-0 class imbalance

Selected endpoints on bus14. Baseline 93.67% survival at a 0.44 action-0 fraction · fixed penalty $\lambda = 0.010 \rightarrow 97.44\%$ at 0.905 · local-safe AIB $d = 0.10$ and $0.20 \rightarrow \approx 98.4\%$ at ≈ 0.98 · evaluation-time p heuristics 97–98% at 0.94–0.97, but imposed after the policy acts.

AIB update. $\lambda_i \leftarrow \text{clip}(\lambda_i + \eta(\hat{C}_i - d), 0, \lambda_{\max})$ after every rollout, with cost weighted by $\sigma(k(p_{\text{safe}} - p_{\text{local}}))$.

Sparse control: the early WCCI stress test

This pilot changes grid, action space and maintenance at the same time — it motivates the graph actor but isolates nothing.



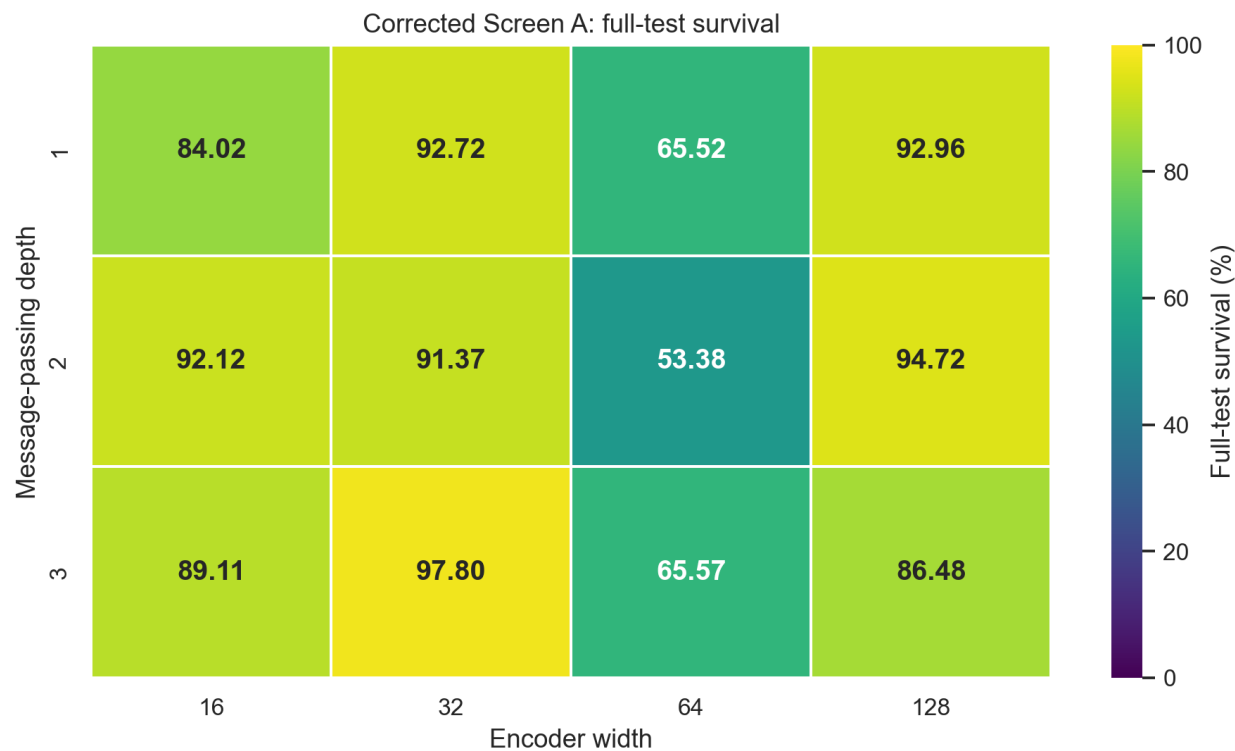
Action-0 / mean-survival trade-off for the maintenance-enabled WCCI pilot. Fixed-width MLP actors, trained from scratch.

Method	Mean survival (%)	Action-0
Matched MLP baseline	23.75 ± 7.70	0.168
Global ρ heuristic, $\rho = 0.90$	28.35 ± 11.06	0.974
Local ρ heuristic, $\rho = 0.90$	9.46 ± 3.54	0.995
Intervention gate, final-action MAP	25.92 ± 10.69	0.400
Flat fixed penalty $\lambda = 0.010$	27.00 ± 12.13	0.489
Flat local-safe AIB, $d = 0.20$	21.65 ± 2.21	0.927

Three seeds, 60M steps, maintenance enabled, no bus14 weights transferred. Values are the final deterministic test evaluation over ten held-out episodes per seed — not full-test estimates. Mechanisms move the intervention frequency but none reliably improves survival, and the differences are small relative to seed variation.

Screen B: encoder depth and width

No monotone rule in depth or width. mp2_h128 is selected on convergence, not on its single best checkpoint.



Full-test mean survival of each run's best checkpoint, seed 0.

Architecture	To 90% (M)	Curve mean (%)	Final-20 mean (%)
mp2_h128	6.22	71.40	94.04
mp1_h16	6.88	70.92	84.93
mp3_h32	7.88	69.63	91.28
mp2_h16	8.79	69.16	88.21
mp1_h128	11.03	68.50	91.11
mp1_h32	6.64	63.13	93.77
mp3_h128	—	58.58	80.20
mp1_h64	—	51.60	63.37
mp3_h64	—	49.80	67.58
mp3_h16	—	48.82	60.00
mp2_h32	13.69	46.51	91.08
mp2_h64	—	45.00	53.41

Convergence measured from the complete training curves. mp3_h32 has the best single checkpoint (97.80%) but a slower, less stable trajectory. Screen A: GINE 94.72% vs GAT 91.06% under a matched busbar/mean-readout setting.

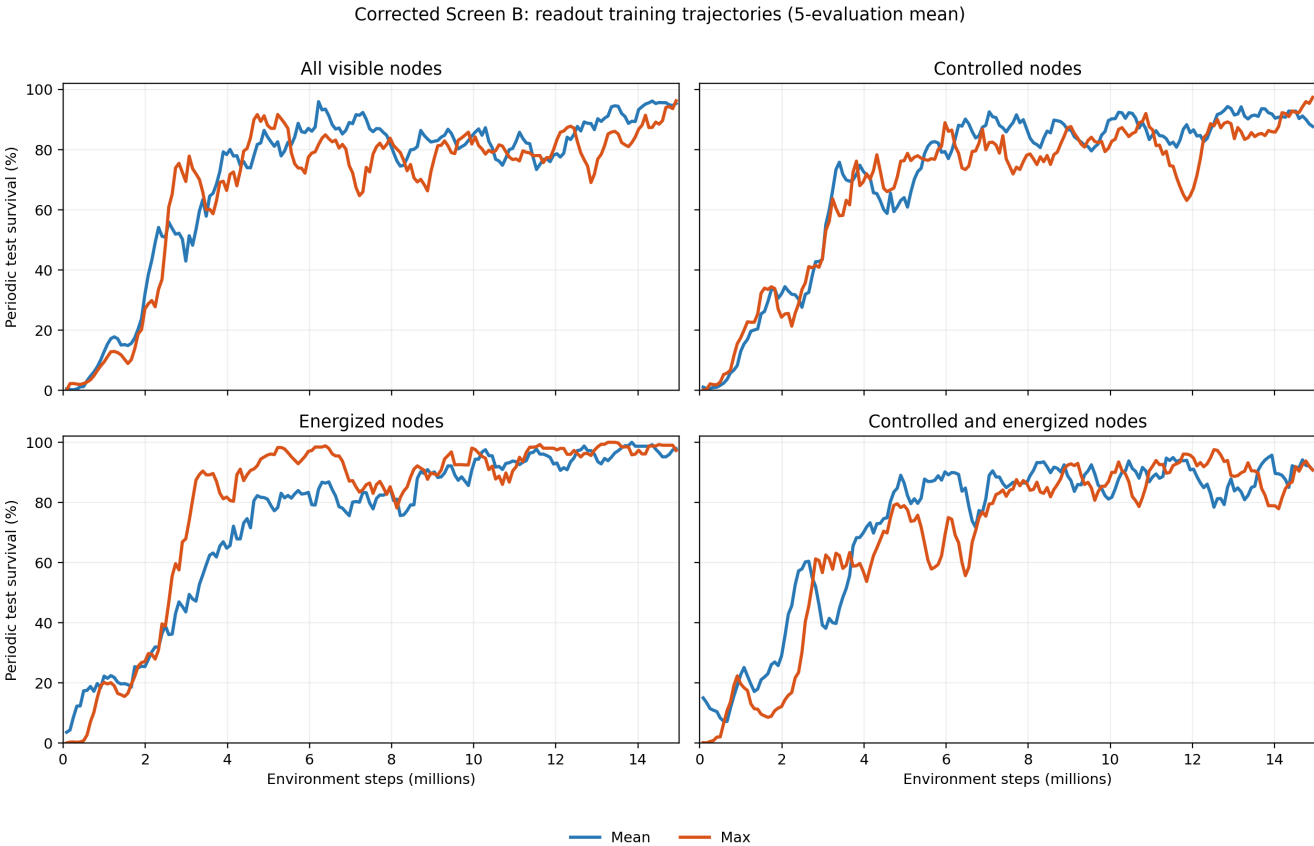
Screen C: graph-readout scope and reducer

Restricting readout to energized nodes is the single largest source-grid gain in the screens.

Readout scope	Mean	Max
All visible	94.72	96.31
Controlled	94.60	94.79
Energized	98.93	99.39
Controlled $\cap$ energized	91.45	96.47

Full-test mean survival (%) by scope and reducer, seed 0. The all-visible/mean cell is inherited from Screen B.

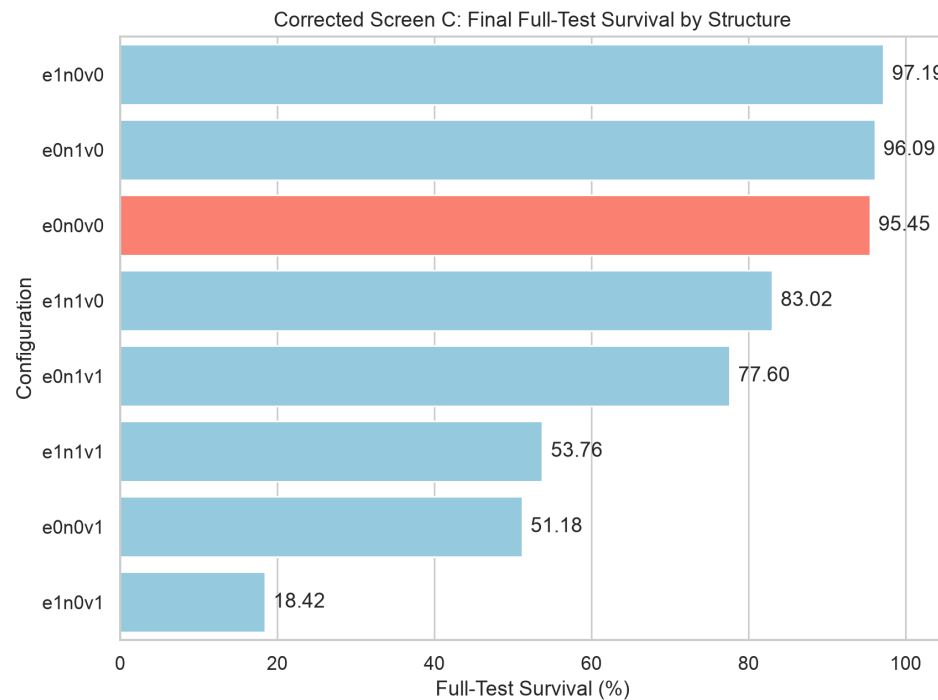
Scope / reducer	To 90% (M)	Curve mean	Final-20 mean	Final-20 std
Energized / max	3.40	79.26	97.79	4.12
Energized / mean	8.71	73.59	98.01	2.90
Ctrl $\cap$ energized / mean	5.97	72.97	91.18	8.59
All visible / mean	6.22	71.40	94.04	5.29
Controlled / mean	6.47	70.94	91.21	6.08
Ctrl $\cap$ energized / max	8.88	68.65	86.31	8.80
All visible / max	4.73	68.38	88.57	8.08
Controlled / max	10.87	68.27	90.67	6.81



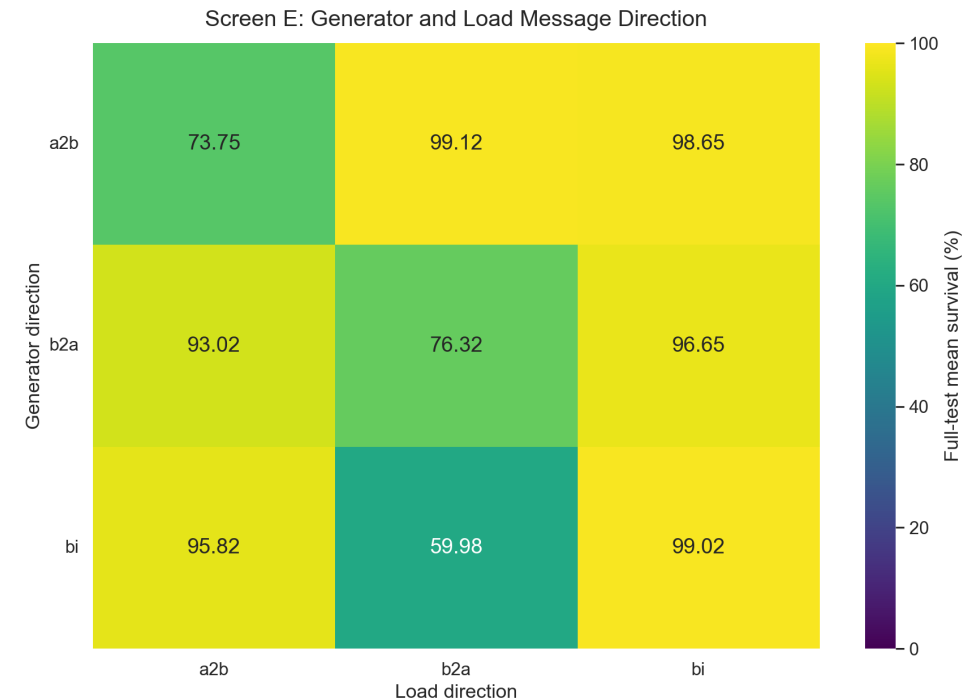
Smoothed training survival by readout scope and reducer.

Screens D and E: augmentations and asset message direction

Structural augmentations do not pay for themselves; asset direction is a coin-flip at one seed.



Screen D. Full 2^3 factorial over same-substation edges (e), substation-summary nodes (n) and virtual-node readout (v). Baseline e0n0v0 highlighted.



Screen E. Full-test survival by generator direction (rows) and load direction (columns) on the disaggregated graph. Best cell gen a2b / load b2a at 99.12%; bi/bi at 99.02%.

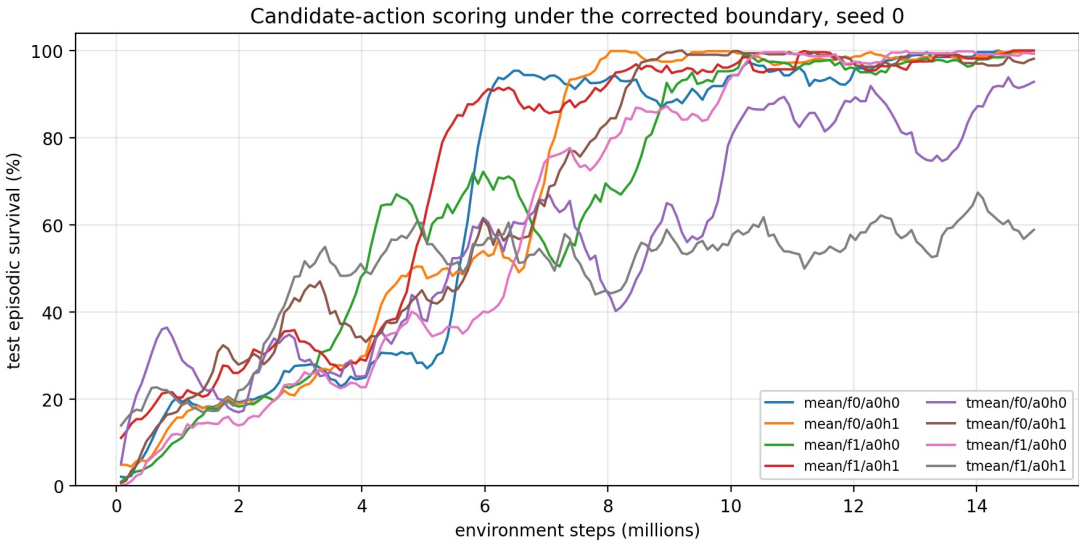
Why virtual-node readout collapses (18–77%). Stacking several aggregation hubs restricts information flow inside a two-layer encoder: the summary and virtual nodes consume hops that the busbars need for physical message passing.

Screen F: the candidate-action head on bus14

Six of eight cells clear 98.2%. No single factor separates the design, so all eight are carried into transfer.

Candidate pooling	Descriptor	Idle head	Mean survival (%)	Best checkpoint
mean	f0	a0h1	99.52	14,929,920
typed_mean	f1	a0h0	99.50	14,929,920
mean	f0	a0h0	99.18	14,929,920
mean	f1	a0h1	98.66	14,929,920
mean	f1	a0h0	98.32	14,846,976
typed_mean	f0	a0h1	98.17	14,764,032
typed_mean	f0	a0h0	92.50	12,358,656
typed_mean	f1	a0h1	57.84	11,612,160

Full-test mean survival on 201 held-out bus14 chronics, one seed per cell.



Training survival for the eight candidate-head cells.

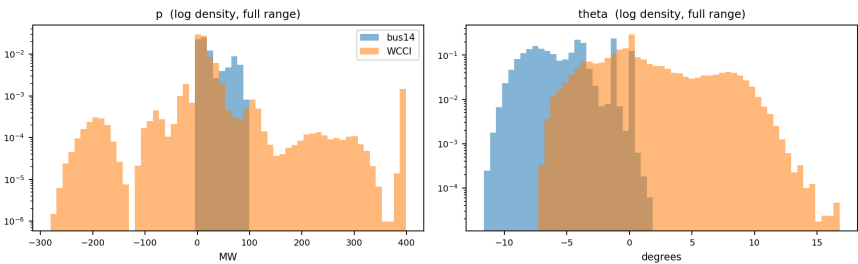
Reading it honestly. The apparent marginal gaps — mean 98.9% against typed_mean 87.0%, f0 97.3% against f1 88.6% — are driven by two low interaction cells. Removing the single 57.84% failure brings the pooling gap to about one point. Typed-mean only shows its value after the grid change.

Cross-grid feature shift: the complete table

Structural zeros hide the mismatch. Only the applicable-row view compares the physical measurements.

Channel	zeros (%)		all sampled rows			applicable rows only		
	zeros bus14	zeros WCCI	offset	ratio	KS	offset	ratio	KS
theta	77.5	73.8	+0.653	0.83	0.157	+1.864	1.43	0.678
p	77.8	76.7	-0.037	2.14	0.042	-0.158	2.54	0.201
rho	63.4	63.6	-0.088	0.92	0.053	-0.260	1.05	0.141
line_status	63.4	63.6	-0.005	1.00	0.002	constant at one		
timestep_overflow	100	100	-0.002	0.70	0.000	-0.004	0.70	0.000
is_busbar	60.6	65.0	-0.093	0.98	0.045	—	—	—
is_generator	91.5	89.3	+0.076	1.11	0.022	—	—	—
is_load	84.5	82.0	+0.066	1.06	0.025	—	—	—
is_transmission_line	63.4	63.6	-0.004	1.00	0.002	—	—	—

Offset in pooled standard deviations; ratio is WCCI ÷ bus14; KS is the maximum ECDF gap. Cooldown channels are constant under do-nothing and omitted from the right block.



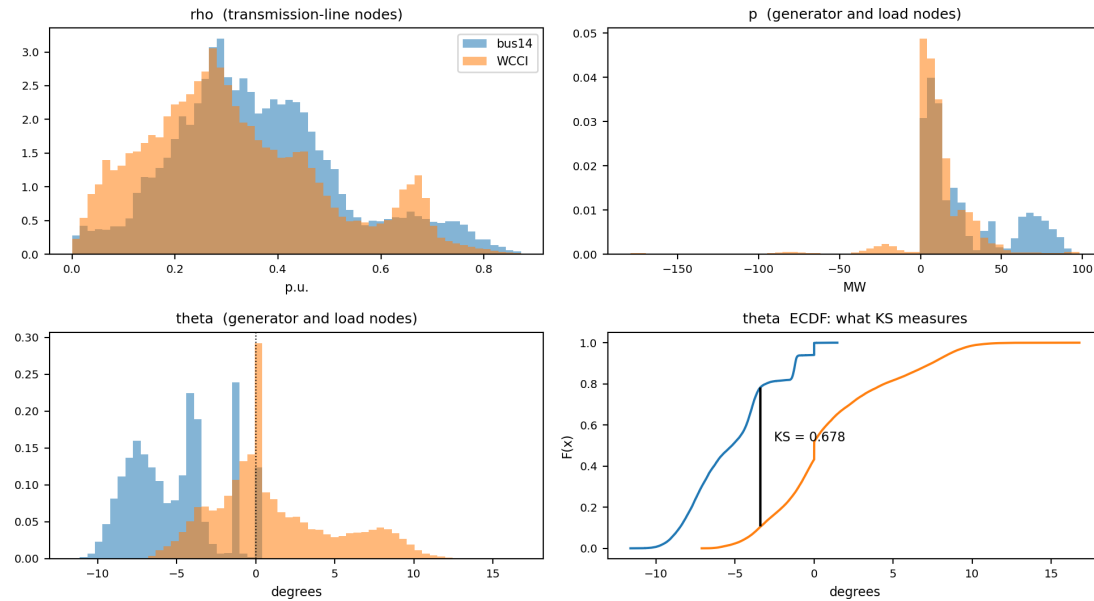
Full range of the two asset channels on a log density scale: WCCI carries a negative net-injection tail absent from bus14.

Divisor	bus14	WCCI	p ratio
Maximum P_max (used)	140	400	0.89
Total P_max	540	2,450	0.56
Generator count	6	22	0.69
Median P_max	85	71	3.04

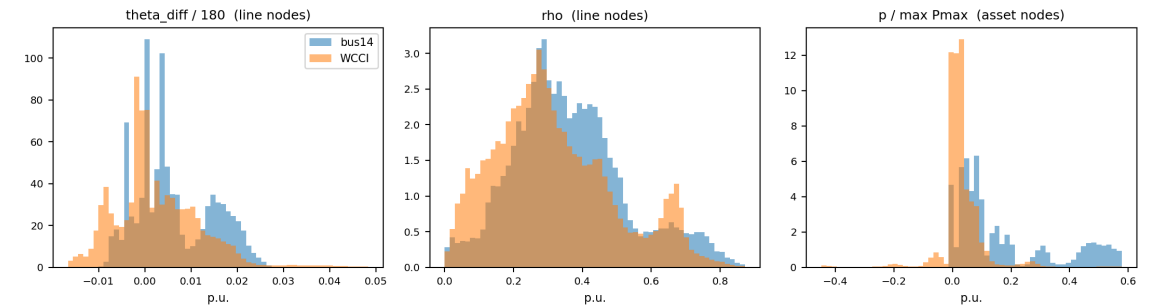
Candidate power normalizers. A ratio of 1.00 is ideal; unscaled it is 2.54.

Feature distributions before and after the transformations

The angle difference behaves like p . Power gains a matched spread but a worse centre.



Raw inputs on the rows where each channel is defined.

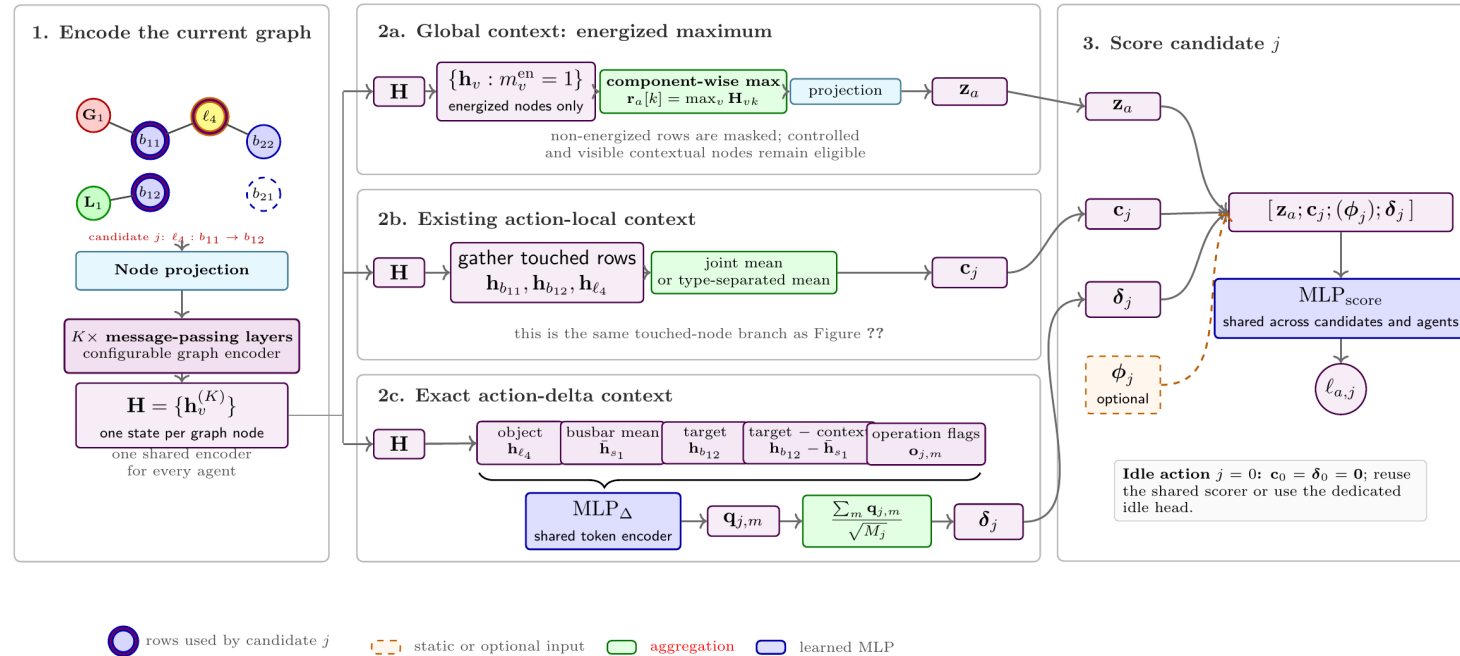


After `gnn_angle_representation = edge_diff` and `gnn_physical_scaling = true`.

Channel	original		KS	transformed		KS
	offset	ratio		offset	ratio	
theta	+1.864	1.43	0.678	replaced		
theta_diff	—	—	—	-0.383	1.15	0.230
p	-0.158	2.54	0.201	-0.808	0.89	0.511
rho	-0.260	1.05	0.141	-0.260	1.05	0.141

The Gmax-delta variant

Same shared scorer, different information: maximum readout plus an explicit encoding of each object modification.



Global vector

z uses a component-wise maximum over energized nodes instead of a mean, so a single stressed node is not averaged away.

Action delta δ

Every exact modification is encoded from the object changed, the mean embedding of its substation busbars, the target busbar, the target-minus-source difference, and static operation flags.

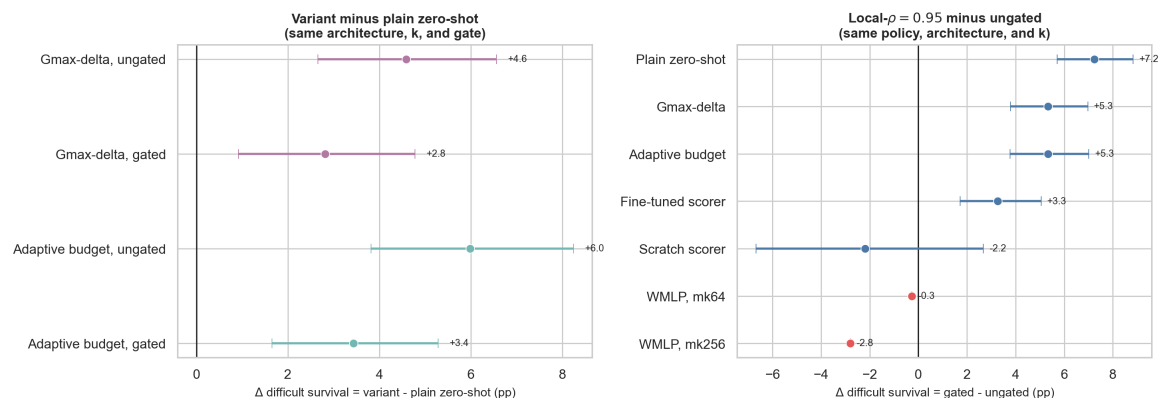
Effect after transfer

+4.6 matched points on the difficult cohort without the gate; median falls from 19.1% at $k = 32$ to 5.1% at $k = 256$, so it does not fix the ranking problem either.

Decision-time changes and the local gate, in full

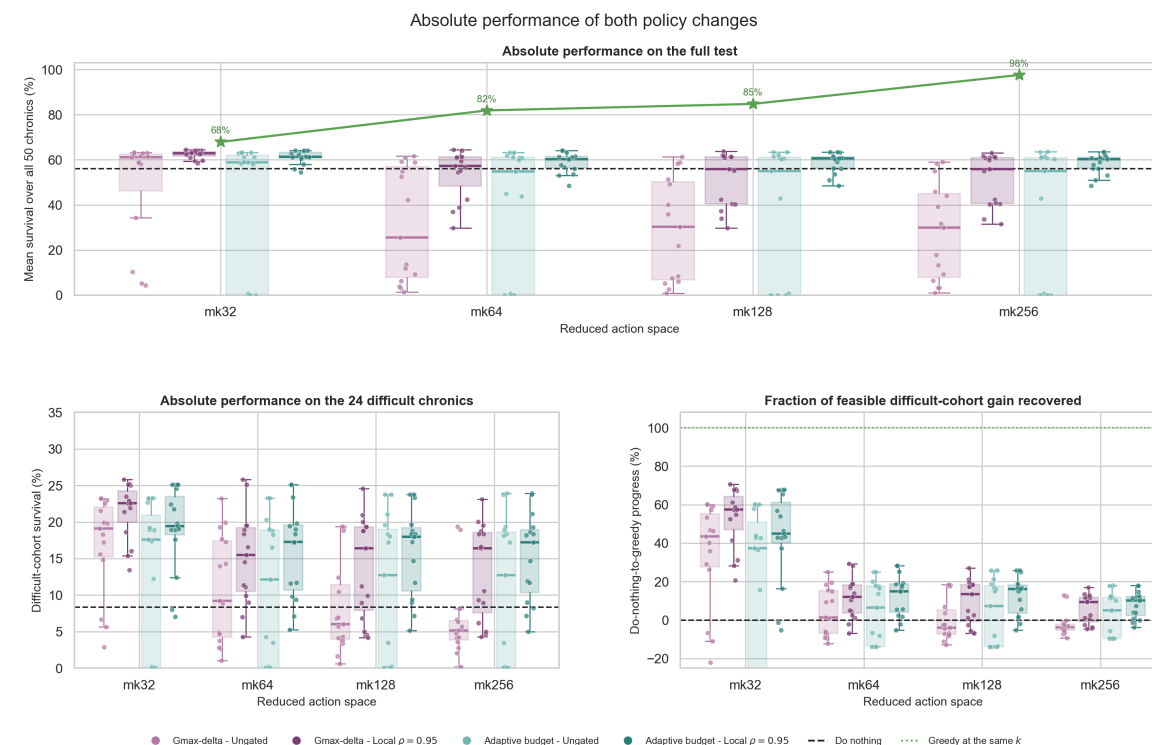
The gate is worth more than either policy change — but only for the transferred actor.

Useful changes are policy-dependent



Matched changes on the difficult cohort. Gmax-delta and the adaptive budget are compared against the same plain zero-shot architectures.

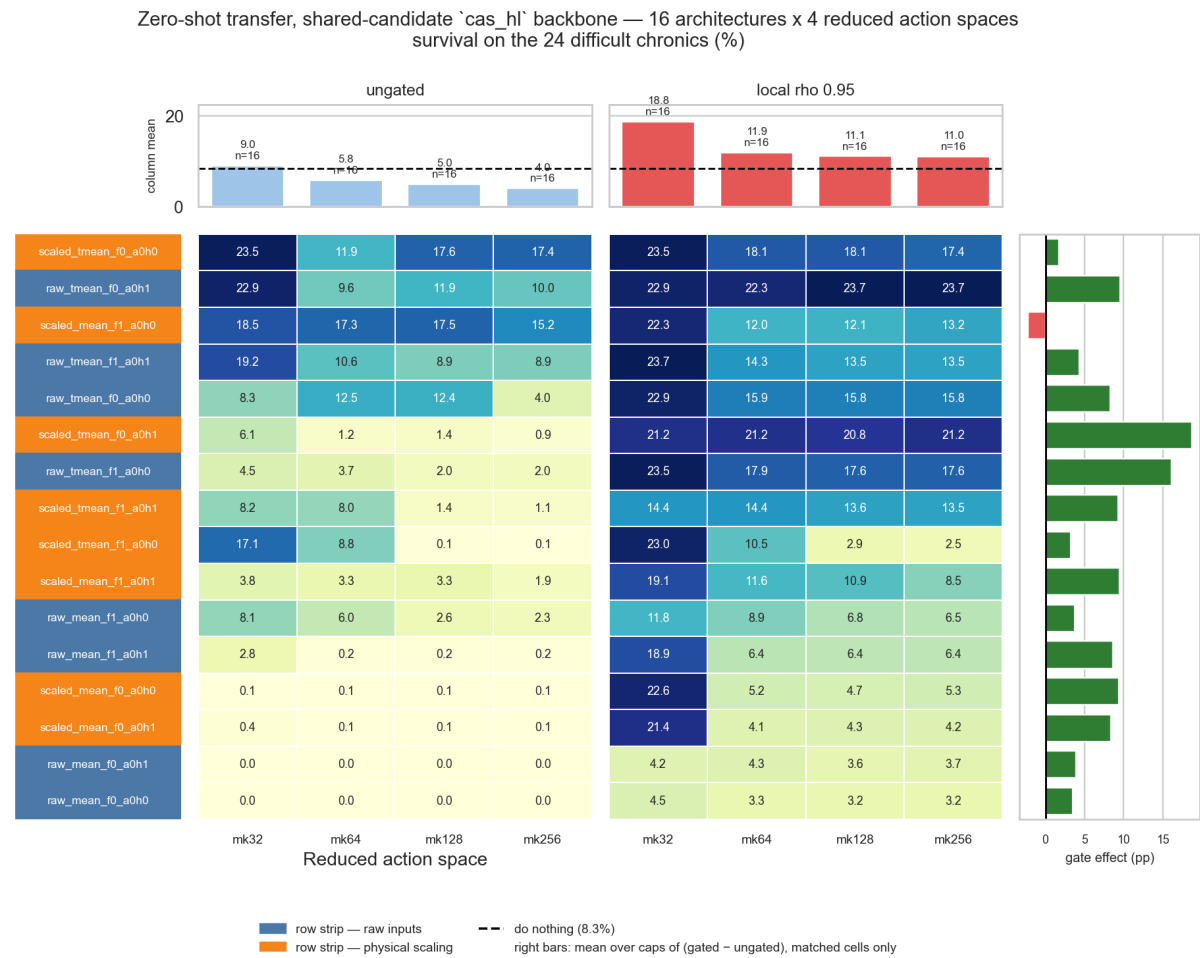
Block sizes. The plain block contains all 64 matched architecture-k pairs; Gmax-delta and the adaptive budget contain 60 each, because their complete designs use 15 architectures rather than 16. The gate improves 58 of 64 plain cells (+7.2 pts) but costs the scratch-trained scorer 2.2 pts and the flat MLP up to 2.8.



Absolute performance at each action-space size: two policy changes × gated and ungated evaluation.

Zero-shot result matrices

Difficult-cohort survival across input condition, candidate pooling, descriptor, idle head and action-space size.



HOW TO READ IT

Rows are architectures, named input · pooling · descriptor · idle head. The coloured strip on the left marks the input condition: blue = raw inputs, orange = physical scaling.

Columns are the four reduced action spaces, given twice — ungated on the left, local-p = 0.95 on the right.

The right-hand bars give the gate effect in points, averaged over caps on matched cells only. Every bar is positive: the gate helps every transferred architecture, though by very different amounts.

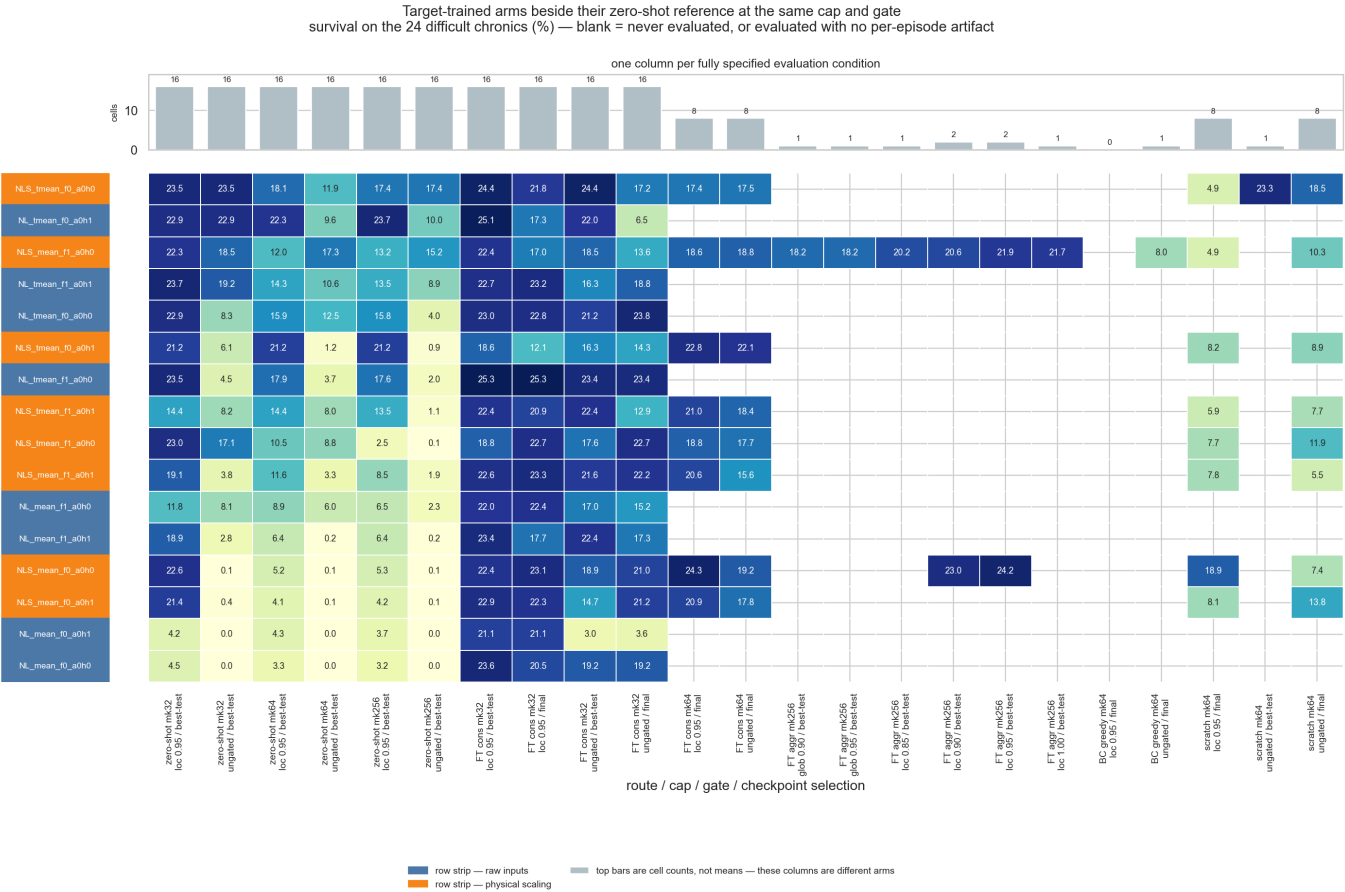
Top bars are column means; the dashed line is do nothing at 8.3%.

Companion matrices for the adaptive intervention budget and the Gmax-delta feature use the same layout and are in the report appendix.

Plain zero-shot actors, all 16 architectures × 4 action-space caps.

Target-trained controls, in detail

The complete board behind the three family means on the scratch-training slide.



PER-AGENT ACTION COUNTS

Space	a0	a1	a2	a3	Total	Greedy
Full	77	65,642	127	1,119	66,965	—
k=32	32	32	32	32	128	33.1%
k=64	64	64	64	64	256	67.9%
k=128	77	128	127	128	460	68.2%
k=256	77	256	127	256	716	94.9%

Greedy is measured on the 24 difficult chronics of the fixed 50-chronic set, not on the full 576-chronic WCCI test split.

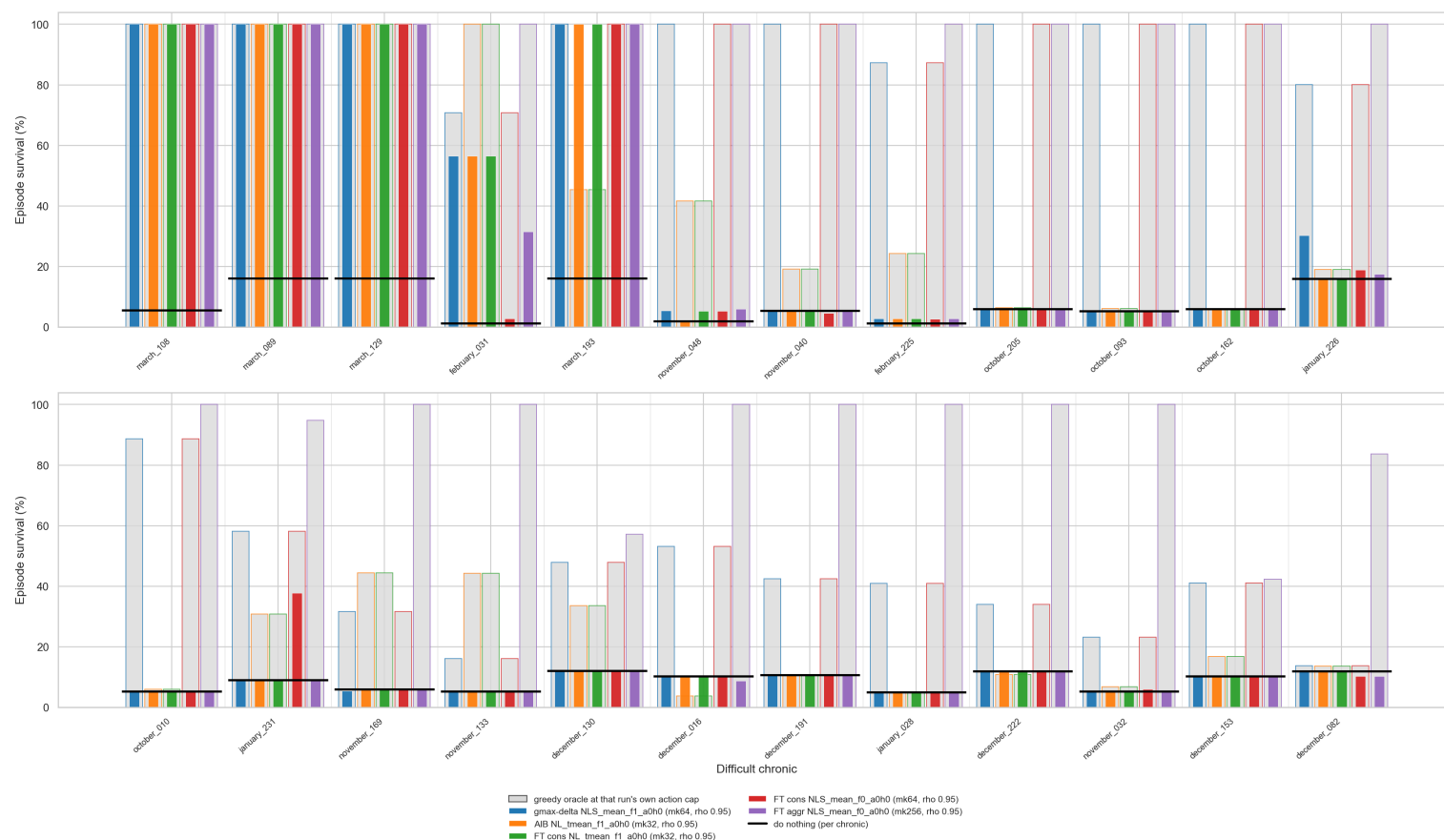
k is a cap, not a fixed size. Agents with fewer valid actions keep their complete list, which is why agent 0 stays at 77 and agent 2 at 127.

Actions are ranked offline before training: states are collected whenever $p_{\max} > 0.95$, each candidate is simulated one step, and an action scores by the fraction of states in which it lowers post-action loading by more than ϵ .

Per-chronic diagnostic on the difficult cohort

Averages hide which specific scenarios a policy rescues and which it breaks.

The 5 strongest runs in the corpus, chronic by chronic, against the greedy oracle at each run's own action cap
wide grey = greedy, narrow colour = the run, black segment = do nothing



The five strongest runs in the corpus, chronic by chronic, against the greedy oracle at each run's own action cap.

HOW TO READ IT

Wide grey is the greedy oracle at that run's own action cap; the narrow coloured bars are the runs; the black segment is do nothing on that chronic.

What it shows. A handful of chronics are solved by almost every policy, and a handful by none. The mean is therefore driven by the middle group, where policies genuinely differ — which is why matched comparisons on the same cohort are more informative than absolute means.

Definitions and likely questions

Chronic One load and generation time series — a scenario. Episode length is up to 8,064 steps.

Mean survival Survived steps $\div$ maximum episode length, averaged over the evaluated chronics.

Easy / difficult cohort Of the fixed 50 WCCI chronics, the 26 that do nothing completes and the 24 it does not.

Rescues The number of difficult chronics a policy completes.

ρ (rho) Line loading divided by that line's own thermal limit. Already grid-relative, which is why it transfers.

k Cap on the number of actions per agent after offline reduction. Agents with fewer valid actions keep their full list.

NL / NLS Original inputs / physical scaling plus line-angle differences.

Typed-mean pooling One mean per node type (bus, line, load, generator), concatenated — width $4 \cdot d_h$ instead of d_h .

Local- ρ gate Block an agent's action while the maximum loading on its own lines is below 0.95.

Greedy progress $100 \cdot (S_{\pi} - S_{DN}) / (S_{\text{greedy},k} - S_{DN})$: 0% is do nothing, 100% is greedy at the same k .

Why difficult-cohort survival and not overall survival? Do nothing already completes 26 of the 50 chronics, so overall survival mostly measures how little damage a policy does to cases that never needed it.